42 topics.
252 passage-based questions.

One passage and six questions for every topic in your requested subject-module list.

Read and reason

Self-contained passages with definitions, formulas, tables and diagrams where relevant.

Choose one answer

Four options per question, labelled a)–d), following the reference’s question structure.

Review and improve

A separate answer key and worked explanations for all 252 questions.

Coverage: mathematics, natural sciences, engineering, computational sciences, business administration, economics, and social sciences and humanities.

Original independent practice material. The requested topics define the coverage; they are not presented as an official exam syllabus. The questions aim at the reasoning level of the reference examples and are not an official or empirically calibrated mock exam.

How to use this question bank

The exercise format

Each exercise begins with a passage, followed by six questions over two pages. Use the passage's definitions and assumptions, including any stated rounding rule. Exactly one option is correct in every question. Questions may test interpretation, calculation or application to a changed situation.

Work from the passage

All information needed for the intended reasoning is provided in the exercise. Treat described studies, businesses and datasets as hypothetical. Unless a question explicitly changes a value or condition, retain those stated in the passage. Diagrams labelled schematic are not to scale.

Practise for speed

First solve a set carefully. Then revisit it under a self-chosen time limit and record both accuracy and time. For each error, identify whether you misread the passage, chose the wrong relationship, made a calculation error or inferred more than the evidence supports.

Use the answers after attempting a set

The compact key is on page 89. Worked explanations begin on page 90. Questions restart at 1 in every exercise; use the exercise number together with the question number when checking answers.

Reference and scope

This bank follows the passage-and-MCQ structure of the Subject Module in the uploaded "dMAT – Preparatory Materials for Test Takers – General Academic Module" (07.07.2026), printed pages 33–57. Its examples are illustrative rather than a complete syllabus or a guarantee of actual test difficulty.

Timing and materials

The reference gives 90 minutes for the full Subject Module and prohibits notes. This 252-question bank is intended for practice across sessions, not as one 90-minute paper. To simulate the stated conditions, work without notes; do not infer additional calculator rules from this bank.

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Vector products, angles, areas and volumes

A design team describes the edges of a three-dimensional frame using vectors. A vector has components and a direction; a scalar has magnitude but no direction. Vector addition and subtraction operate component by component. The length of u = (ux, uy, uz) is √(ux² + uy² + uz²).

The dot product u · v = ux vx + uy vy + uz vz is a scalar. For nonzero vectors it also equals |u||v| cos θ, where θ is the angle between them. The cross product is u × v = (uy vz - uz vy, uz vx - ux vz, ux vy - uy vx). It is a vector perpendicular to both original vectors when the product is nonzero. Reversing the order reverses its sign.

The magnitude |u × v| gives the area of their parallelogram. The absolute scalar triple product |u · (v × w)| gives the volume of the parallelepiped. A zero triple product means the three vectors are coplanar.

For this frame, use a = (2, 0, 0), b = (1, 3, 0) and c = (0, 0, 4).

Question 1

What is a · b?

a) 2

b) 8

c) 0

d) 6

Question 2

Which expression gives the angle between a and b?

a) arccos(0)

b) arccos(1/3)

c) arccos(1/√10)

d) arccos(3/√10)

Vector products, angles, areas and volumes

Use the passage and any figures or tables on the preceding page.

Question 3

Which vector equals a × b?

a) (0, 6, 0)
b) (2, 3, 0)
c) (0, 0, -6)
d) (0, 0, 6)

Question 4

What is the area of the parallelogram spanned by a and b?

a) 12 square units
b) 2√10 square units
c) 6 square units
d) 3 square units

Question 5

What is the volume spanned by a, b and c?

a) 24 cubic units
b) 12 cubic units
c) 48 cubic units
d) 6 cubic units

Question 6

If c is replaced by (3, 1, 0), which conclusion follows?

a) Every vector must have zero length.
b) The vectors a and b become perpendicular.
c) The volume is zero because all three vectors lie in the same plane.
d) The area spanned by a and b becomes zero.

Systems of linear equations and algebra

A laboratory buys two types of reusable container. Let x and y denote their quantities. The order records give x + y = 14 and 2x + y = 20. A third quantity is defined as z = 3x - y. All quantities in this exercise are treated as real numbers during algebraic manipulation; the resulting container counts must be nonnegative integers.

An equation remains equivalent when the same number is added to both sides or both sides are multiplied by the same nonzero number. Substitution replaces a variable with an equal expression. Elimination combines equations to remove a variable; for example, subtracting one complete equation from another subtracts both left sides and both right sides.

A system has a unique solution when its independent constraints identify one combination of values. Two equations can instead represent the same constraint, giving no new information, or contradict one another, giving no solution. In questions about changed records, change only the condition explicitly mentioned and retain the other relevant equation.

Question 1

What is x for the original order?

a) 6
b) 8
c) 14
d) 20

Question 2

What is z for the original order?

a) 10
b) 2
c) 26
d) 18

Systems of linear equations and algebra

Use the passage and any figures or tables on the preceding page.

Question 3

Which operation eliminates y immediately?

a) Multiply both equations by 2.
b) Subtract the first equation from the second.
c) Replace y with 14 in both equations.
d) Add the two original equations.

Question 4

If 2x + y = 22 replaces the second record, while x + y = 14 remains, how does x change?

a) It decreases by 2.
b) It increases by 2.
c) It stays unchanged.
d) It increases by 1.

Question 5

Which equation is equivalent to x + y = 14?

a) 2x + y = 28
b) x + 2y = 14
c) 2x + 2y = 28
d) 2x + 2y = 14

Question 6

What happens if x + y = 14 and 2x + 2y = 30 are imposed together?

a) There is exactly one solution.
b) The only solution is x = y = 15.
c) There is no solution.
d) Every pair with x + y = 14 is a solution.

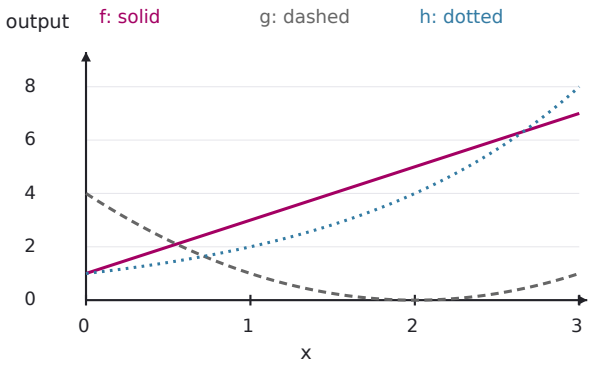
Functions and graphs

A monitoring device uses four functions. The input x is dimensionless. The linear function f(x) = 2x + 1 changes by the same amount for every unit increase in x. The quadratic function g(x) = (x - 2)² has a lowest point, called its vertex. The exponential function h(x) = 2^x multiplies its output by 2 when x increases by 1. The logarithmic function k(x) = $\log_2(x)$, defined only for x > 0, is the inverse of h: k(x) = y means $2^y = x$.

Adding a constant to an entire function shifts every output by that constant without changing the input. A point is an intersection of two graphs when both functions have the same output at the same input. The table lists selected outputs; it is not a claim that intersections can occur only at integer inputs.

Use the equations whenever an input is absent from the table. An expression such as $\log_2(8)$ asks for an exponent rather than multiplication by 2.

x	f(x)	g(x)	h(x)
0	1	4	1
1	3	1	2
2	5	0	4
3	7	1	8



Question 1

Which function has a constant rate of change with respect to x?

a) g
b) f
c) k
d) h

Question 2

At which input does g reach its minimum?

a) x = 1
b) x = 0
c) x = 2
d) x = 4

Functions and graphs

Use the passage and any figures or tables on the preceding page.

Question 3

What is h(3)?

a) 6
b) 3
c) 8
d) 9

Question 4

What is k(8)?

a) 1/3
b) 4
c) 3
d) 16

Question 5

The graph of f is shifted upward by 3 units. Which function results?

a) 2(x + 3) + 1
b) 2x + 4
c) 2x - 2
d) 5x + 1

Question 6

Among the four inputs shown in the table, where do f and h have equal outputs?

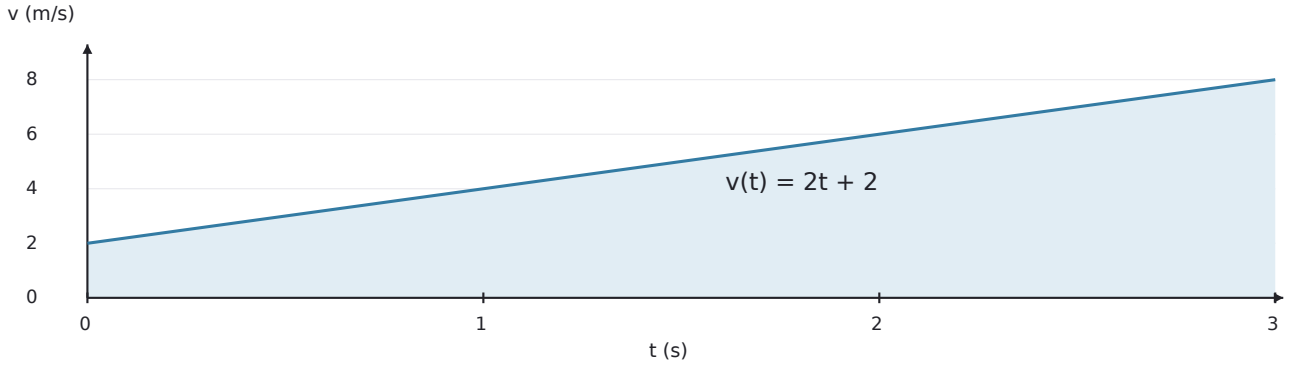
a) x = 1
b) x = 0
c) x = 2
d) x = 3

Differentiation and integration

A trolley moves along a straight track. Between t = 0 and t = 3 seconds, its position in metres is s(t) = 5 + t² + 2t. Positive position is measured in a fixed direction from a reference point. The instantaneous velocity is the derivative v(t) = ds/dt. The acceleration is dv/dt. The derivative rules needed here are d(t²)/dt = 2t, d(kt)/dt = k for constant k, and the derivative of a constant is zero.

Average velocity over an interval is the change in position divided by elapsed time. It need not equal the instantaneous velocity at either endpoint. The definite integral of velocity gives displacement: ∫ from a to b v(t) dt = s(b) - s(a). Thus signed area under a velocity-time graph represents displacement. When velocity is nonnegative throughout an interval, displacement also equals the distance travelled.

An integral can be evaluated using an antiderivative: ∫(2t + 2) dt = t² + 2t + constant. Apply the upper endpoint and subtract the lower endpoint.



Question 1

What is the instantaneous velocity at t = 2 s?

a) 4 m/s
b) 8 m/s
c) 13 m/s
d) 6 m/s

Question 2

What is the displacement from t = 0 to t = 3 s?

a) 5 m
b) 9 m
c) 15 m
d) 20 m

Differentiation and integration

Use the passage and any figures or tables on the preceding page.

Question 3

What is the average velocity over the same three seconds?

a) 5 m/s
b) 15 m/s
c) 20/3 m/s
d) 6 m/s

Question 4

If the constant 5 in s(t) is replaced by 12, what happens to velocity?

a) It is unchanged at every t.
b) It doubles at every t.
c) It increases by 7 m/s at every t.
d) It becomes zero at t = 0.

Question 5

Which expression correctly represents the distance travelled in this interval?

a) ∫ from 0 to 3 (5 + t² + 2t) dt
b) ∫ from 0 to 3 (2t + 2) dt
c) s(3) + s(0)
d) v(3) - v(0)

Question 6

Which statement about the acceleration is correct?

a) It is zero because the path is straight.
b) It is constant at 5 m/s².
c) It is constant at 2 m/s².
d) It increases as 2t + 2 m/s².

Probability and descriptive statistics

A sensor produces the four possible readings 2, 4, 4 and 6, each of the four listed outcomes being equally likely. Treat these four outcomes as the entire population. The population mean μ is the sum of the readings divided by their number. The population variance is the mean of the squared deviations from μ, and the standard deviation is the square root of the variance. Use division by 4 here, not the sample-variance denominator 3.

Adding a constant to every reading shifts the mean by that constant but does not change the deviations from the mean. Multiplying all readings by a constant multiplies the standard deviation by its absolute value.

For a separate sensor, assume readings follow a normal distribution with mean 10 and standard deviation 2. Approximately 68% of its readings lie within one standard deviation of its mean. Finally, two independent quality checks each fail with probability 0.2. Independence means the probability that both fail is the product of the individual probabilities.

Question 1

What is the mean of the four equally likely outcomes?

a) 3
b) 16
c) 4
d) 8

Question 2

What are their population variance and standard deviation?

a) Variance 4; standard deviation 2
b) Variance 2; standard deviation 2
c) Variance 2; standard deviation √2
d) Variance 8; standard deviation √8

Probability and descriptive statistics

Use the passage and any figures or tables on the preceding page.

Question 3

Approximately what percentage of readings from the separate normal sensor lie between 8 and 12?

a) 100%
b) 95%
c) 68%
d) 34%

Question 4

What is the probability that both independent quality checks fail?

a) 0.4
b) 0.04
c) 0.8
d) 0.2

Question 5

If 3 is added to every original reading, what changes?

a) The mean becomes 7 and the variance becomes 5.
b) The mean becomes 12 and the variance becomes 6.
c) The mean remains 4 and the variance becomes 5.
d) The mean becomes 7 and the variance remains 2.

Question 6

For the original four equally likely outcomes, what is P(reading ≥ 4)?

a) 1
b) 1/2
c) 3/4
d) 1/4

Matrices and determinants

A two-channel device transforms an input column vector by matrix multiplication. Its matrix is M = [[2, 1], [1, 1]]. For an input (x, y), the output is (2x + y, x + y). The first row therefore determines the first output and the second row determines the second.

For a 2 × 2 matrix [[a, b], [c, d]], the determinant is ad - bc. If the determinant is nonzero, the matrix has an inverse and each output corresponds to exactly one input. The inverse is (1/(ad - bc)) × [[d, -b], [-c, a]]. A zero determinant means an inverse does not exist; depending on the requested output, the corresponding equations may have no solution or more than one.

Multiplying one row by a factor multiplies the determinant by that factor. Matrix products apply transformations in order: MN means apply N first and then M to a column vector. For comparison, let N = [[1, 0], [0, 2]], which leaves the first component unchanged and doubles the second.

Question 1

What output does M produce from input (3, 2)?

a) (5, 8)

b) (6, 2)

c) (7, 6)

d) (8, 5)

Question 2

What is det(M)?

a) 1

b) 3

c) 0

d) -1

Matrices and determinants

Use the passage and any figures or tables on the preceding page.

Question 3

Which matrix is M inverse?

a) [[1, -1], [-1, 2]]
b) [[2, -1], [-1, 1]]
c) [[1, 1], [1, 2]]
d) [[2, 1], [1, 1]]

Question 4

What follows for the matrix [[2, 4], [1, 2]]?

a) It has an inverse because all entries are positive.
b) Every output has exactly one corresponding input.
c) It maps every input to the zero vector.
d) It has no inverse because its determinant is zero.

Question 5

If only the first row of M is doubled, what is the new determinant?

a) 4
b) 0
c) 2
d) 1

Question 6

Which statement about MN and NM is correct?

a) MN = [[2, 2], [1, 2]], while NM = [[2, 1], [2, 2]].
b) Both products equal [[2, 2], [1, 2]].
c) MN and NM are undefined because the matrices are square.
d) Both products equal [[2, 1], [2, 2]].

Kinematics and Newton’s laws

A 2 kg trolley starts from rest on a horizontal track. During its first four seconds, a constant horizontal pulling force of 8 N acts to the right and a constant resisting force of 2 N acts to the left. Treat the trolley as a particle. Its weight and the upward support force balance, so they do not affect horizontal acceleration.

Newton’s second law is Fnet = ma, where Fnet is the vector sum of the forces on the same object. For constant acceleration, velocity is v = u + at and displacement is s = ut + ½at². Here u is initial velocity and t is elapsed time. A zero net force means constant velocity, which may be zero or nonzero; it does not necessarily mean the object is at rest.

Newton’s third law states that interacting objects exert equal and opposite forces on one another. These two forces act on different objects, so they must not be added as though both act on the trolley. Where a question changes the mass, assume the stated horizontal forces are maintained.

Question 1

What is the horizontal acceleration during the first four seconds?

a) 3 m/s²

b) 6 m/s²

c) 5 m/s²

d) 4 m/s²

Question 2

What is the velocity after four seconds?

a) 6 m/s

b) 8 m/s

c) 24 m/s

d) 12 m/s

Kinematics and Newton’s laws

Use the passage and any figures or tables on the preceding page.

Question 3

How far does the trolley travel in those four seconds?

a) 12 m
b) 6 m
c) 48 m
d) 24 m

Question 4

Why do the pulling force on the trolley and its third-law partner not cancel in the trolley’s force sum?

a) The partner force acts on the pulling object.
b) The partner force acts only when the trolley stops.
c) The partner force has no direction.
d) The partner force is smaller than the pulling force.

Question 5

After four seconds, both horizontal forces are removed. What happens on an otherwise resistance-free track?

a) The trolley reverses its direction.
b) The trolley continues at constant nonzero velocity.
c) The trolley stops immediately.
d) The trolley continues with its previous acceleration.

Question 6

If the mass were 4 kg and both original forces stayed the same, what would the acceleration be?

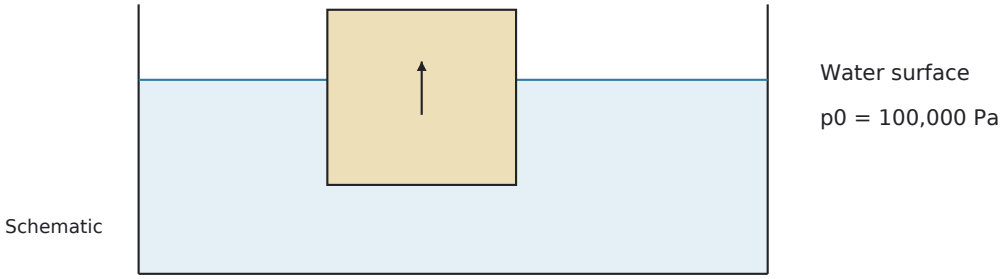
a) 6 m/s²
b) 2 m/s²
c) 1.5 m/s²
d) 3 m/s²

Hydrostatics, pressure and buoyancy

A tank contains fresh water at rest and is open to air. Use water density ρ = 1,000 kg/m³, gravitational acceleration g = 10 m/s², and surface pressure p0 = 100,000 Pa. Absolute pressure at vertical depth h is p = p0 + ρgh. Gauge pressure is the excess above the surface pressure, namely ρgh. One bar equals 100,000 Pa. At a given depth in the same stationary liquid, pressure is independent of the tank's shape.

The buoyant force on an object is ρgVsub, where Vsub is its submerged volume. A freely floating object is in vertical equilibrium when this equals its weight mg. Consider a rigid block with total volume 0.020 m³ and mass 12 kg. Ignore surface tension and any external supporting force.

A separate flexible, sealed air pocket has volume 2.0 L at an absolute pressure of 1.0 bar. At constant temperature and with a fixed amount of air, use pV = constant. This pocket is separate from the rigid block: changes to the pocket must not be applied to the block's fixed volume.



Schematic

Question 1

What is the absolute water pressure at a depth of 5 m?

a) 600,000 Pa
b) 50,000 Pa
c) 150,000 Pa
d) 100,000 Pa

Question 2

What is the gauge pressure at that depth?

a) 200,000 Pa
b) 150,000 Pa
c) 50,000 Pa
d) 10,000 Pa

Hydrostatics, pressure and buoyancy

Use the passage and any figures or tables on the preceding page.

Question 3

What fraction of the rigid block’s volume is submerged when it floats?

a) 0.40
b) 1.00
c) 0.60
d) 0.12

Question 4

What is the greatest buoyant force on the block when it is completely submerged in this water?

a) 20 N
b) 200 N
c) 320 N
d) 120 N

Question 5

What is the air pocket’s volume at an absolute pressure of 2.0 bar, with temperature unchanged?

a) 1.0 L
b) 4.0 L
c) 2.0 L
d) 0.5 L

Question 6

The rigid block is moved to a liquid of density 800 kg/m³. What submerged volume is required for floating?

a) 0.0096 m³
b) 0.020 m³
c) 0.015 m³
d) 0.012 m³

Heat, temperature and the ideal gas law

Temperature describes a thermal state; heat is energy transferred because of a temperature difference. When no phase change occurs and specific heat capacity is constant, heating a body requires $Q = mc\Delta T$. Use m in kilograms, c in J/(kg·K), and temperature change ΔT in kelvin. A temperature change of 1 K has the same size as a change of 1 °C. For absolute gas temperatures in this exercise, use $T(K) = T(°C) + 273$.

A 2 kg liquid sample has $c = 4{,}000$ J/(kg·K). Assume all supplied heat enters the sample and no phase change occurs. A separate ideal-gas system obeys $pV = nRT$. For a fixed amount of gas, R and n are constant. Ratios involving this law require absolute pressure and temperature in kelvin.

For the gas, the first law is $\Delta U = Q - W$. Here Q is positive for heat entering the gas and W is positive for work done by the gas on its surroundings. An insulated process has $Q = 0$, but its internal energy can still change if work is done.

Question 1

How much does the liquid sample’s temperature rise after receiving 80,000 J?

a) 20 K
b) 40 K
c) 5 K
d) 10 K

Question 2

At fixed volume and fixed gas amount, temperature rises from 300 K to 600 K. What happens to pressure?

a) It halves.
b) It doubles.
c) It remains unchanged.
d) It increases by a factor of four.

Heat, temperature and the ideal gas law

Use the passage and any figures or tables on the preceding page.

Question 3

A gas receives 500 J of heat and does 200 J of work. What is ΔU?

a) 300 J
b) 700 J
c) -300 J
d) -700 J

Question 4

An insulated gas does 100 J of work on its surroundings. What is ΔU?

a) 200 J
b) 0 J
c) -100 J
d) 100 J

Question 5

Which temperature should replace 27 °C in the ideal gas law under the stated approximation?

a) 27 K
b) 327 K
c) 246 K
d) 300 K

Question 6

At constant pressure, a fixed gas amount occupies 2 L at 300 K. What volume does it occupy at 450 K?

a) 4 L
b) 1.33 L
c) 2 L
d) 3 L

Electrical quantities, Ohm’s law and power

A test bench contains ideal direct-current voltage sources and resistors whose resistance remains constant. For a resistor, Ohm’s law is V = IR: V is the potential difference in volts, I the current in amperes, and R the resistance in ohms. Electrical power is P = VI, which can also be written I^2R or V^2/R. Energy transferred during a constant-power interval is E = Pt.

A watt is one joule per second. A watt-hour is the energy delivered by one watt over one hour; 1 kilowatt-hour equals 1,000 watt-hours. Do not confuse an energy value with a power value.

Unless a question explicitly changes the arrangement, the bench has a single 6 Ω resistor connected across an ideal 12 V source. Neglect wire resistance and changes caused by heating. When resistors are connected in series, the same current passes through each and their resistances add. The source maintains its stated terminal voltage even when the connected resistance changes.

Question 1

What current flows through the original resistor?

a) 6 A

b) 2 A

c) 72 A

d) 0.5 A

Question 2

What power does the original resistor dissipate?

a) 72 W

b) 12 W

c) 24 W

d) 2 W

Electrical quantities, Ohm’s law and power

Use the passage and any figures or tables on the preceding page.

Question 3

How much energy is dissipated in five seconds?

a) 5 J
b) 120 J
c) 4.8 J
d) 24 J

Question 4

If the voltage across the same resistor doubles, by what factor does its power change?

a) 1
b) 1/2
c) 2
d) 4

Question 5

The 6 Ω resistor is replaced by 2 Ω and 4 Ω resistors in series across 12 V. What current passes through the 4 Ω resistor?

a) 3 A
b) 6 A
c) 2 A
d) 1 A

Question 6

How much energy does the original resistor use in two hours, expressed in kWh?

a) 0.48 kWh
b) 0.024 kWh
c) 48 kWh
d) 0.048 kWh

Stoichiometry, concentration and pH

Chemical equations express ratios of reacting amounts in moles. Consider $2H_2 + O_2 \rightarrow 2H_2O$. Complete reaction consumes two moles of hydrogen for each mole of oxygen and produces two moles of water. When reactants are not present in the required ratio, the limiting reactant is consumed first and determines the maximum product amount. Assume complete reaction and no side reactions in this exercise. The starting amounts are 6 mol H_2 and 2 mol O_2.

For a separate dissolved substance, molar concentration is $c = n/V$, where n is solute amount in moles and V is total solution volume in litres. During dilution with pure solvent, the solute amount is unchanged, so $c_1V_1 = c_2V_2$. Use final solution volume, not merely the amount of solvent added.

For the idealised dilute solutions used here, take $pH = -\log_{10}[H^+]$, where the numerical hydrogen-ion concentration is expressed in mol/L. Since $\log_{10}(10^x) = x$, multiplying $[H^+]$ by ten reduces pH by one. The reaction, dilution and pH scenarios are independent.

Question 1

Which starting reactant limits water production?

a) Oxygen

b) Both are exhausted simultaneously.

c) Hydrogen

d) Neither can be limiting.

Question 2

How many moles of water can form?

a) 8 mol

b) 2 mol

c) 6 mol

d) 4 mol

Stoichiometry, concentration and pH

Use the passage and any figures or tables on the preceding page.

Question 3

What is the concentration of 0.50 mol solute in 2.0 L of solution?

a) 1.0 mol/L
b) 4.0 mol/L
c) 0.25 mol/L
d) 0.50 mol/L

Question 4

A 0.20 L sample at 0.50 mol/L is diluted to a final volume of 1.0 L. What is the final concentration?

a) 2.50 mol/L
b) 0.50 mol/L
c) 0.10 mol/L
d) 0.125 mol/L

Question 5

What is the pH when $[H^+] = 10^{-3}$ mol/L?

a) 11
b) -3
c) 0.001
d) 3

Question 6

What happens to pH if $[H^+]$ increases from 10^{-3} to 10^{-2} mol/L?

a) It remains at 3.
b) It increases from 3 to 4.
c) It increases from 3 to 30.
d) It decreases from 3 to 2.

Waves and mechanical energy

A travelling periodic wave is described by frequency f, wavelength λ and speed v, related by v = fλ. Frequency counts oscillations per second and is measured in hertz. Wavelength is the distance between equivalent points on successive cycles. In the medium considered here, wave speed is fixed, so changing frequency changes wavelength inversely. For this particular wave model at fixed frequency, transported energy is proportional to the square of amplitude.

For a separate moving object, kinetic energy is Ek = ½mv² and gravitational potential energy near Earth is Ep = mgh, relative to a chosen height zero. Use g = 10 m/s². If only gravity does work, total mechanical energy Ek + Ep is conserved. If friction converts some mechanical energy into heat, mechanical energy alone decreases, although total energy is conserved when the thermal change is included.

Unless a question states otherwise, consider a 2 kg object and ignore air resistance. A drop begins from rest. The wave and falling-object examples are independent and the symbol v denotes the relevant speed in each.

Question 1

A wave has frequency 5 Hz and wavelength 2 m. What is its speed?

a) 2.5 m/s

b) 10 m/s

c) 0.4 m/s

d) 7 m/s

Question 2

What is the kinetic energy of the 2 kg object moving at 3 m/s?

a) 9 J

b) 6 J

c) 18 J

d) 3 J

Waves and mechanical energy

Use the passage and any figures or tables on the preceding page.

Question 3

What potential energy does the object have at a height of 5 m above the reference level?

a) 100 J
b) 10 J
c) 25 J
d) 50 J

Question 4

After dropping from rest through 5 m without resistance, what is its speed?

a) 100 m/s
b) 20 m/s
c) 5 m/s
d) 10 m/s

Question 5

At fixed frequency, doubling the wave amplitude multiplies the transported energy by what factor?

a) 4
b) 2
c) 1
d) 1/2

Question 6

If wave frequency doubles while wave speed stays fixed, what happens to wavelength?

a) It stays unchanged.
b) It halves.
c) It increases fourfold.
d) It doubles.

Statics: forces, moments and equilibrium

A light horizontal beam is supported at its left end L and right end R, which are 4 m apart. Treat its own weight as negligible. A downward 60 N load acts 1 m from L, and a downward 40 N load acts 3 m from L. The supports provide vertical upward reactions RL and RR. The diagram shows the original arrangement.

A body is in static equilibrium when the net force and net moment are both zero. A force's moment about a point equals its magnitude times the perpendicular distance from that point to its line of action. Opposite rotation senses have opposite signs. Taking moments about L removes RL because its lever arm is zero: $4RR = 60 \times 1 + 40 \times 3$. Vertical force balance gives $RL + RR = 60 + 40$.

Equal and opposite parallel forces on different lines of action can create a couple: their net force is zero but their moment is not. The couple magnitude is one force multiplied by the perpendicular separation between the two force lines. Ignore support deformation.



Question 1

What is the sum RL + RR for the original arrangement?

a) 60 N
b) 100 N
c) 180 N
d) 40 N

Question 2

What is the reaction RR at the right support?

a) 45 N
b) 55 N
c) 25 N
d) 75 N

Statics: forces, moments and equilibrium

Use the passage and any figures or tables on the preceding page.

Question 3

The 60 N load moves from 1 m to 2 m from L. How much does RR increase?

a) 60 N
b) 0 N
c) 15 N
d) 30 N

Question 4

A 10 N force acts with a perpendicular lever arm of 2 m. What is its moment magnitude?

a) 12 N·m
b) 20 N·m
c) 40 N·m
d) 5 N·m

Question 5

Why is zero net force alone insufficient to establish static equilibrium?

a) A nonzero net moment can still cause rotational acceleration.
b) Moment balance replaces the need for force balance.
c) A zero net force always means that every individual force is zero.
d) A zero net force requires a nonzero net moment.

Question 6

Two opposite parallel 20 N forces have lines of action separated by 0.5 m. What couple magnitude do they produce?

a) 40 N·m
b) 0 N·m
c) 20 N·m
d) 10 N·m

Stress, strain and basic beam behaviour

A straight uniform bar is stretched by a central axial tensile force. Engineering stress is $\sigma = F/A$, where A is its original cross-sectional area. Engineering strain is $\varepsilon = \Delta L/L$, where L is original length and ΔL is elongation. Within the linear elastic range, $\sigma = E\varepsilon$, with E the material's Young's modulus. Strain is dimensionless. A stress of 1 N/mm² equals 1 MPa, and 1,000 MPa equals 1 GPa.

For the original bar, F = 2,000 N, A = 20 mm², L = 1,000 mm and ΔL = 0.50 mm. Assume uniform stress, small deformation and unchanged material properties. The linear elastic relation must not automatically be extended past the material's elastic range, where permanent deformation can occur.

For a separate simply supported beam with a central load, a small-deflection model gives $\delta = FL^3/(48EI)$. Here I is the cross-section's second moment of area, not electrical current. In scaling questions, keep all quantities fixed except the one explicitly changed. This model shows how beam length and stiffness affect deflection without requiring the full bending derivation.

Question 1

What is the original bar's tensile stress?

a) 100 MPa
b) 10 MPa
c) 1,000 MPa
d) 40 MPa

Question 2

What is the original bar's strain?

a) 0.0005
b) 0.005
c) 0.5
d) 500

Stress, strain and basic beam behaviour

Use the passage and any figures or tables on the preceding page.

Question 3

What Young’s modulus follows from these measurements?

a) 200 GPa
b) 0.2 GPa
c) 2 GPa
d) 20 GPa

Question 4

If the bar’s area doubles with force, length and material unchanged, what happens to elongation in the linear range?

a) It halves.
b) It remains unchanged.
c) It doubles.
d) It becomes four times as large.

Question 5

In the stated beam model, doubling L while holding F, E and I fixed changes deflection by what factor?

a) 4
b) 2
c) 1/2
d) 8

Question 6

Which situation most directly prevents automatic use of $\sigma = E\varepsilon$ for the bar?

a) The force is applied along the bar’s axis.
b) The original length is expressed in millimetres.
c) The cross-sectional area is constant.
d) The loading has caused deformation beyond the linear elastic range.

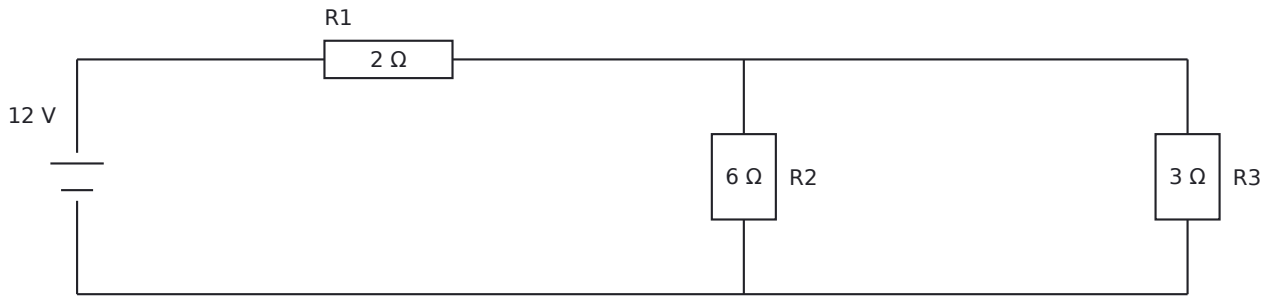
Series, parallel and Kirchhoff’s laws

An ideal 12 V source supplies a 2 Ω resistor R1 in series with a parallel pair: R2 = 6 Ω and R3 = 3 Ω. Neglect wire resistance, source internal resistance and temperature changes. The circuit diagram shows which components share the same two junctions.

Series components carry the same current and their resistances add. Parallel branches have the same voltage across them. Their equivalent resistance Rp satisfies 1/Rp = 1/R2 + 1/R3. For each resistor, V = IR.

Kirchhoff’s current law states that current entering a junction equals current leaving it, expressing charge conservation in the steady state. Kirchhoff’s voltage law states that the signed voltage changes around a closed loop sum to zero. In this circuit, the voltage across R1 plus the voltage across either complete parallel branch equals 12 V.

If R3 is removed, assume that only its branch becomes open. R2 remains connected, so the circuit becomes R1 and R2 in series. Do not treat removal of one parallel branch as a break in every path through the circuit.



Question 1

What is the equivalent resistance of R2 and R3 together?

a) 2 Ω
b) 9 Ω
c) 3 Ω
d) 4.5 Ω

Question 2

What total current does the source supply in the original circuit?

a) 6 A
b) 3 A
c) 1.5 A
d) 2 A

Series, parallel and Kirchhoff’s laws

Use the passage and any figures or tables on the preceding page.

Question 3

What voltage is across R2?

a) 6 V
b) 9 V
c) 3 V
d) 12 V

Question 4

What current passes through R3 in the original circuit?

a) 2 A
b) 1 A
c) 4 A
d) 3 A

Question 5

After R3 is removed as described, what current does the source supply?

a) 6 A
b) 3 A
c) 1.5 A
d) 0 A

Question 6

Which equation expresses current conservation at the original branch junction?

a) $3\ \text{A} = 1\ \text{A} + 2\ \text{A}$
b) $2\ \text{A} \times 3\ \Omega = 6\ \text{V}$
c) $6\ \Omega + 3\ \Omega = 9\ \Omega$
d) $12\ \text{V} = 6\ \text{V} + 6\ \text{V}$

Energy conversion and efficiency

A two-stage lifting system converts electrical energy into useful mechanical energy. Stage 1 is a motor; stage 2 is a transmission. Efficiency is useful output energy divided by input energy, using the same units. For steady operation, the same ratio may be calculated using powers. Efficiencies are expressed either as fractions or percentages.

During one cycle, 1,000 J enters the motor, 800 J leaves it as mechanical energy, and 600 J reaches the lifted load after the transmission. Treat all energy that does not reach the named useful output as a loss from that stage. It is converted into other forms rather than destroyed. No additional energy enters between stages.

For sequential stages, overall efficiency is the product of stage efficiencies. The input to stage 2 is the useful output from stage 1, not the original system input. Improvements in one stage do not automatically alter the other stage’s efficiency. When a question specifies a new steady power or a separate required output, apply the same ratio definition to those stated quantities.

Question 1

What is the motor’s efficiency in the original cycle?

a) 60%
b) 20%
c) 75%
d) 80%

Question 2

What is the transmission’s efficiency?

a) 75%
b) 25%
c) 60%
d) 80%

Energy conversion and efficiency

Use the passage and any figures or tables on the preceding page.

Question 3

What is the overall useful-energy efficiency?

a) 15%
b) 77.5%
c) 60%
d) 155%

Question 4

How much input energy does not reach the load as useful mechanical energy?

a) 800 J
b) 400 J
c) 600 J
d) 200 J

Question 5

If transmission efficiency improves to 90% while motor efficiency and the 1,000 J input remain unchanged, what reaches the load?

a) 800 J
b) 900 J
c) 750 J
d) 720 J

Question 6

A separate converter receives 500 W and supplies 300 W of useful power steadily. What is its efficiency?

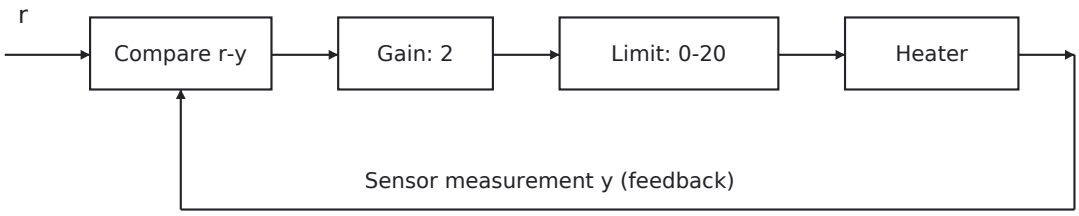
a) 200%
b) 40%
c) 167%
d) 60%

Control and feedback loops

A heater controller compares a target temperature r with a measured temperature y. Its error is e = r - y. A proportional controller computes a command u = 2e, but the actuator limits the actual command to the range 0 to 20 command units: values above 20 are clipped to 20, and negative values are clipped to zero. Larger positive commands supply more heating. Initially, r = 60 °C and the sensor reports y = 52 °C.

This is negative feedback: if the measured temperature falls below the target, the controller increases heating. A disturbance such as additional heat loss can change the temperature even if the target remains constant. An open-loop heater instead applies a predetermined command without using the measured temperature to adjust it.

A sensor bias means the reported temperature differs systematically from the actual temperature. The controller acts on the report. A larger proportional gain produces a larger unsaturated command for the same error; it does not by itself guarantee stable operation, particularly when response delays matter. Do not assume this proportional controller always reaches zero steady-state error.



Question 1

What is the initial error e?

a) 8 °C
b) 4 °C
c) 112 °C
d) -8 °C

Question 2

What actual command is sent initially?

a) 8 units
b) 16 units
c) 20 units
d) 4 units

Control and feedback loops

Use the passage and any figures or tables on the preceding page.

Question 3

The target changes to 55 °C while the sensor still reads 52 °C. What command results?

a) 0 units
b) 10 units
c) 6 units
d) 16 units

Question 4

If a sensor reads exactly 5 °C too high, what is the actual temperature when its display reads 60 °C?

a) 60 °C
b) 65 °C
c) 55 °C
d) 52 °C

Question 5

Which change would make the described heating strategy open-loop?

a) Subtract the measurement from the target before computing the command.
b) Apply a fixed command without using the temperature measurement.
c) Increase the command when a measurement falls.
d) Limit the feedback command to 20 units.

Question 6

With an error of 15 °C, what actual command is sent?

a) 20 units
b) 15 units
c) 30 units
d) 0 units

Unit conversions and dimensional analysis

A laboratory combines measurements from instruments using different units. Conversions must preserve the physical quantity, and powers of length require powers of the conversion factor. Use 1 km = 1,000 m, 1 h = 3,600 s, 1 cm = 0.01 m, 1 g = 0.001 kg and 1 kWh = 3.6×10^6 J. Therefore 1 $cm^2 = 10^{-4}$ m^2 and 1 $cm^3 = 10^{-6}$ m^3.

Density is mass divided by volume. Pressure is force divided by area, with 1 Pa = 1 N/m^2. The newton has dimensions $kg \cdot m/s^2$. Speed has dimensions m/s, and acceleration has dimensions m/s^2.

Dimensional analysis checks whether the units on both sides of a proposed relationship agree. Terms added together must have the same dimensions. Matching dimensions is necessary for a physical formula, but does not prove that its numerical coefficient or underlying model is correct. In the questions, treat symbols as physical quantities with the dimensions stated; do not treat them as unit-free numbers merely because their numerical values are simple.

Question 1

What is 72 km/h in m/s?

a) 20 m/s

b) 72 m/s

c) 7.2 m/s

d) 200 m/s

Question 2

What is 2 g/cm^3 in kg/m^3?

a) 200 kg/m^3

b) 0.002 kg/m^3

c) 2,000 kg/m^3

d) 20 kg/m^3

Unit conversions and dimensional analysis

Use the passage and any figures or tables on the preceding page.

Question 3

What is 3 cm^2 in m^2?

a) 0.003 m^2
b) 0.0003 m^2
c) 0.000003 m^2
d) 0.03 m^2

Question 4

A 100 N force is distributed uniformly over 2 cm^2. What is the pressure?

a) 500,000 Pa
b) 50,000 Pa
c) 50 Pa
d) 5,000 Pa

Question 5

If v is speed and g is acceleration, which expression has dimensions of length?

a) vg
b) v/g
c) g/v^2
d) v^2/g

Question 6

What is 0.5 kWh in joules?

a) 1,800,000 J
b) 180,000 J
c) 7,200,000 J
d) 1,800 J

Binary and hexadecimal number systems

A controller stores nonnegative integers using positional number systems. In base 2, the digits are 0 and 1; the place values from right to left are 1, 2, 4, 8, 16 and so on. For example, 101_2 represents $1 \times 4 + 0 \times 2 + 1 \times 1$. In base 16, place values are powers of 16. The digits A, B, C, D, E and F represent decimal values 10 through 15. A subscript identifies the base; a number without a subscript is decimal.

Four binary digits correspond to one hexadecimal digit. To group a binary number into hexadecimal digits, form groups of four from the right and add leading zeros if needed. Leading zeros do not change the value.

For separate register questions, use an unsigned four-bit register. It holds values from 0 to 15. Arithmetic wraps modulo 16: any carry beyond the fourth bit is discarded. A left shift by one moves every bit one place left, inserts zero at the right, and discards a bit that moves beyond the left boundary.

Question 1

What is 1101_2 in decimal?

a) 11

b) 13

c) 14

d) 9

Question 2

Which binary numeral represents decimal 26?

a) 11100_2

b) 10011_2

c) 10110_2

d) 11010_2

Binary and hexadecimal number systems

Use the passage and any figures or tables on the preceding page.

Question 3

What is $2F_{16}$ in decimal?

a) 215
b) 32
c) 31
d) 47

Question 4

Which hexadecimal numeral equals 101000_2?

a) 18_{16}
b) $A0_{16}$
c) 40_{16}
d) 28_{16}

Question 5

What is stored after adding 1 to 1111 in the specified four-bit register?

a) 0000
b) 10000
c) 0001
d) 1111

Question 6

What is stored after shifting 0011 left by one bit in that register?

a) 1100
b) 0110
c) 0010
d) 0001

Boolean logic and logic gates

A safety indicator uses Boolean inputs A, B and C, each either 0 or 1. A means that the indicator is enabled; B and C are two sensor signals. The output is X = A AND (B OR C). AND gives 1 only when both inputs are 1; OR gives 1 when at least one input is 1; NOT reverses 0 and 1. A separate XOR gate gives 1 when its two inputs differ, and 0 when they are the same.

Boolean operations describe truth values, not ordinary arithmetic addition. Parentheses specify which intermediate operation is evaluated first. Two useful identities are NOT(B OR C) = (NOT B) AND (NOT C), and A AND (B OR C) = (A AND B) OR (A AND C). An equivalent circuit produces the same output for every possible input combination.

There are eight possible combinations of the three inputs because each input has two choices. Unless a question changes the formula, use the indicator rule X = A AND (B OR C). The indicator is a simplified logical model, with no delay or hardware faults.

Question 1

What is X when A = 1, B = 0 and C = 1?

a) The output is not determined.

b) 2

c) 1

d) 0

Question 2

If A = 0, what can be concluded regardless of B and C?

a) X equals B XOR C.

b) X is always 0.

c) X is always 1.

d) X equals B OR C.

Boolean logic and logic gates

Use the passage and any figures or tables on the preceding page.

Question 3

Which expression is equivalent to NOT(B OR C)?

a) (NOT B) AND (NOT C)
b) (NOT B) OR (NOT C)
c) NOT(B AND C)
d) B AND C

Question 4

What does a separate XOR gate output when both its inputs are 1?

a) 0
b) 1
c) It alternates between 0 and 1.
d) 2

Question 5

Which circuit has the same output as the indicator for every input combination?

a) OR of A, B and C
b) OR of the outputs of (A AND B) and (A AND C)
c) AND of A, B and C
d) AND of the outputs of (A OR B) and (A OR C)

Question 6

For how many of the eight input combinations does X equal 1?

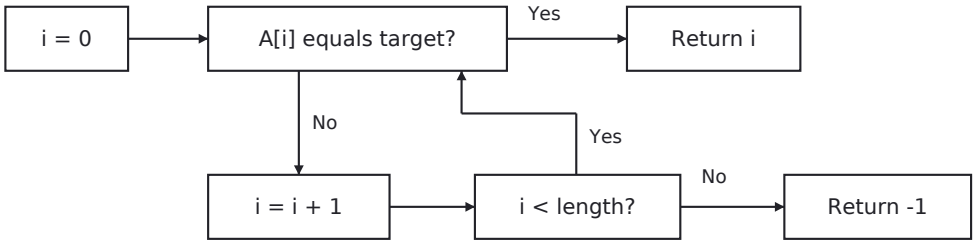
a) 3
b) 4
c) 1
d) 7

Algorithms and step-by-step tracing

An algorithm is a specified sequence of steps. A linear-search routine scans an array from left to right, comparing each element with a target. It returns the index of the first match and stops; if no match exists, it returns -1 after checking every element. Array indices start at 0. The original array is [7, 2, 9, 4].

A separate sorting routine makes one left-to-right pass. It compares the elements at positions 0 and 1 and swaps them if the left is larger; it then does the same for positions 1 and 2, and finally positions 2 and 3. Each comparison uses the current array, including any earlier swaps. One pass is not necessarily a complete sort.

Binary search instead requires ascending sorted data. It checks index floor((low + high)/2), where low and high initially mark the first and last indices. If the middle value is below the target, it continues in the upper half; if above, in the lower half; if equal, it returns that index. Count only comparisons between a target and an array element when asked for search comparisons.



Question 1

What index does linear search return for target 9 in the original array?

a) 3
b) 2
c) -1
d) 1

Question 2

How many element comparisons does linear search make for target 8 in the original array?

a) 1
b) 2
c) 3
d) 4

Algorithms and step-by-step tracing

Use the passage and any figures or tables on the preceding page.

Question 3

What array results from one complete sorting pass as defined?

a) [7, 2, 4, 9]
b) [2, 7, 4, 9]
c) [2, 9, 7, 4]
d) [2, 4, 7, 9]

Question 4

Which property is required before using the stated binary-search procedure?

a) The array must be sorted in ascending order.
b) Every element must be a positive integer.
c) The array must contain an even number of elements.
d) The array must contain no repeated values.

Question 5

For sorted array [2, 4, 7, 9] and target 4, what happens at the first binary-search comparison?

a) Index 1 is checked and the target is found.
b) Index 2 is checked and the target is found.
c) Index 0 is checked and the lower half is discarded.
d) Index 3 is checked and the search reports absence.

Question 6

What does linear search return for target 4 in [4, 2, 4]?

a) 1
b) -1
c) 2
d) 0

Algorithmic complexity and growth

An operation-count model estimates how work changes with input size n. Consider three algorithms with counts TA(n) = 3n + 5, TB(n) = n^2 + n, and TC(n) = ceil($\log_2$ n) + 1 for positive integers n. The ceiling operation rounds a value upward to the nearest integer. A logarithmic count commonly arises when the remaining search region is repeatedly halved.

Big-O notation describes an asymptotic upper growth bound, not an exact runtime or a guarantee about which implementation is faster at every small input. In these questions, “growth class” means the tightest class among the options that describes the dominant growth of the given count. Ignore fixed machine-dependent time per operation.

A constant number of sequential linear passes still has linear growth. Two nested loops, each executing n iterations for every iteration of the other, perform n^2 inner operations. Sequential stages add their work; the fastest-growing stage dominates the overall asymptotic class. For example, a sorting stage of growth n log n followed by a single logarithmic search is dominated by the sort.

Question 1

What is the growth class of TA(n)?

a) $O(n^2)$

b) O(n)

c) O(log n)

d) O(1)

Question 2

For very large n, doubling n changes TB(n) by approximately what factor?

a) 8

b) 2

c) 1

d) 4

Algorithmic complexity and growth

Use the passage and any figures or tables on the preceding page.

Question 3

What is TC(8)?

a) 3
b) 8
c) 4
d) 9

Question 4

Two fully nested loops each run n iterations. What is the growth class of the inner operation count?

a) O(n)
b) $O(n^2)$
c) O(1)
d) O(log n)

Question 5

What is the combined growth class of a sort of O(n log n) followed by one O(log n) search?

a) O(n)
b) $O(n^2 \log n)$
c) O(log n)
d) O(n log n)

Question 6

Which statement correctly compares A and B at n = 1?

a) B has fewer modelled operations even though its asymptotic growth is larger.
b) A and B have equal operation counts.
c) Their counts cannot be calculated from the supplied formulas.
d) A has fewer modelled operations because linear growth always wins.

Arrays, stacks and queues

Data structures differ in the order in which stored items can be accessed. An array stores an ordered sequence and supports access by position; use zero-based indexing, so the first item has index 0. For these exercises, reading an array item does not remove it.

A stack follows last-in, first-out order. “Push x” places x on its top, and “pop” removes and returns the top item. A queue follows first-in, first-out order. “Enqueue x” adds x at its back, and “dequeue” removes and returns the front item. Every operation in a question occurs in the written order, and a stated empty structure initially contains no items.

Attempting to remove an item from an empty stack or queue causes an underflow error in this model; it does not return an invented value. A printer that must process jobs in arrival order can use a queue. An undo mechanism that must reverse the most recent action first can use a stack. The choice follows the required access order rather than the visual arrangement of items on a screen.

Question 1

An empty stack receives push A, push B, then pop. What is returned?

a) Both A and B

b) B

c) An underflow error

d) A

Question 2

An empty queue receives enqueue A, enqueue B, then dequeue. What is returned?

a) B

b) Both A and B

c) An underflow error

d) A

Arrays, stacks and queues

Use the passage and any figures or tables on the preceding page.

Question 3

For array [9, 5, 7, 1], what value is read at index 2?

a) 7
b) 9
c) 1
d) 5

Question 4

An empty stack receives push 3, push 5, pop, push 7, pop. What does the final pop return?

a) 5
b) 3
c) 7
d) An underflow error

Question 5

Which pairing matches the required access order?

a) Both tasks require a queue.
b) Printer jobs in arrival order: stack; undo latest action: queue.
c) Both tasks require a stack.
d) Printer jobs in arrival order: queue; undo latest action: stack.

Question 6

An empty queue receives enqueue X, dequeue, dequeue. What happens at the last operation?

a) The value 0 is returned.
b) A new X is inserted automatically.
c) An underflow error occurs.
d) X is returned a second time.

Recursion and iteration

A recursive function solves a problem using a call to itself on a smaller input. A base case stops further calls. Consider F on nonnegative integers: F(0) = 1; for n > 0, F(n) = n + F(n - 1). Each call must wait for the smaller call’s value before completing the addition. Count the initial call and the base-case call when asked how many calls occur.

An iterative procedure starts with x = 1 and, for each integer i from 1 through n inclusive, replaces x with x + i. It then returns x. “Inclusive” means the update for i = n is performed. For n = 0 there are no loop updates and the procedure returns 1.

A second recursive function G has G(n) = 1 for n ≤ 1, and G(n) = 2 + G(n - 2) for n > 1. Use only positive integer starting inputs for G here. Recursion and iteration can compute the same values, but termination depends on the specified stopping condition and on progress toward it; calling the same unchanged positive input would not reach F’s base case.

Question 1

What is F(3)?

a) 4

b) 3

c) 6

d) 7

Question 2

Why does F terminate for every nonnegative integer input in its stated domain?

a) Every recursive call reduces n by 1 until n = 0.

b) The recursive rule applies only when n = 0.

c) Every recursive call doubles n.

d) The input is never changed.

Recursion and iteration

Use the passage and any figures or tables on the preceding page.

Question 3

What does the iterative procedure return when n = 3?

a) 4
b) 6
c) 7
d) 1

Question 4

How many total calls occur when evaluating F(4), including the initial and base-case calls?

a) 4
b) 8
c) 5
d) 3

Question 5

What is G(5)?

a) 3
b) 5
c) 7
d) 6

Question 6

The iterative loop for n = 3 is changed to process only i < 3, starting at i = 1. What value is returned?

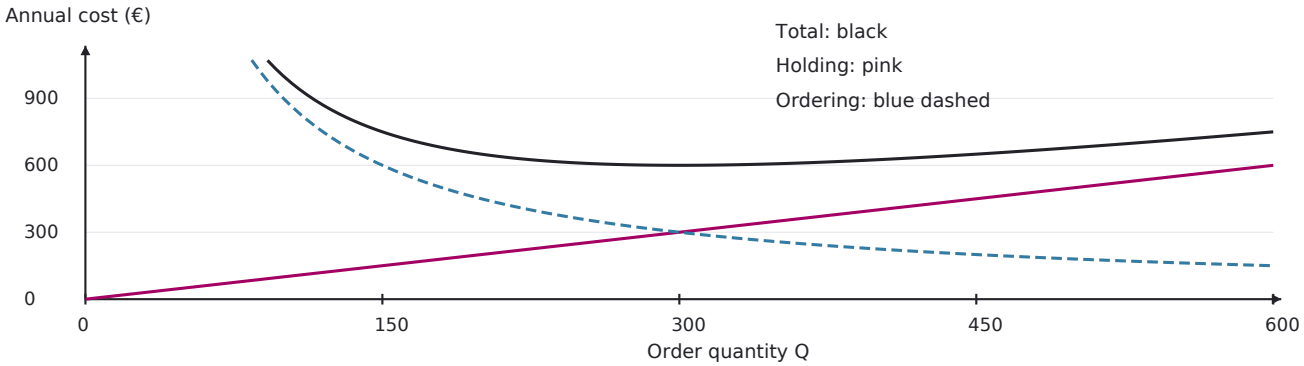
a) 7
b) 3
c) 4
d) 6

Optimal order quantity and inventory costs

A retailer uses the basic economic order quantity model. Annual demand D is known and constant, each order incurs a fixed ordering cost S, and holding one unit for a year costs H. Replenishment is instantaneous, shortages do not occur, and there are no quantity discounts, storage limits or financing limits. Inventory falls steadily from order quantity Q to zero before the next delivery, so average inventory is Q/2.

Annual ordering cost is DS/Q and annual holding cost is HQ/2. Their sum is the relevant cost C(Q) = DS/Q + HQ/2. Purchase cost per unit is constant and therefore does not change the choice of Q. The continuous optimum is Q* = √(2DS/H). At this optimum, the two relevant annual cost components are equal.

Use D = 3,600 units per year, S = €25 per order and H = €2 per unit per year unless a question changes one parameter. A sensitivity question changes only the named input. The model selects an order size, not a reorder date, and a change that breaks an assumption may require a different cost model.



Question 1

What is the optimal order quantity?

a) 150 units
b) 900 units
c) 600 units
d) 300 units

Question 2

What is average inventory when orders contain 300 units?

a) 75 units
b) 150 units
c) 12 units
d) 300 units

Optimal order quantity and inventory costs

Use the passage and any figures or tables on the preceding page.

Question 3

How many orders are placed per year when Q = 300?

a) 300
b) 24
c) 12
d) 6

Question 4

If the fixed cost per order doubles while D and H stay the same, how does Q* change?

a) It is multiplied by √2.
b) It halves.
c) It is divided by √2.
d) It doubles.

Question 5

What is the combined annual ordering and holding cost at Q = 300?

a) €600
b) €7,200
c) €300
d) €900

Question 6

Which change most directly invalidates using the stated basic model without modification?

a) Holding cost is expressed in euros per unit per year.
b) Annual demand remains known and constant.
c) The supplier introduces a quantity discount that changes unit purchase cost.
d) The retailer orders the same quantity each time.

Break-even analysis

A workshop sells one product at price p per unit. It incurs fixed operating costs F during the period and variable cost v for each unit sold. Assume every unit produced is sold, price and unit variable cost remain constant, and there are no other costs. Revenue is pq, total cost is F + vq, and operating profit is π = (p - v)q - F, where q is the number of units sold.

The contribution margin per unit is p - v: it is the amount each sale contributes toward fixed costs and then profit. When p > v, the break-even quantity is qBE = F/(p - v). If indivisible units produce a fractional break-even result, the minimum whole-unit quantity avoiding a loss is rounded upward.

For the original workshop, F = €12,000, p = €50 and v = €30. Each question that changes price, cost or capacity is a separate scenario starting from these original figures. Capacity is the maximum number of units that can be sold in the period and does not itself alter the fixed cost or unit margin.

Question 1

What is the original contribution margin per unit?

a) €80
b) €30
c) €50
d) €20

Question 2

What is the original break-even quantity?

a) 400 units
b) 600 units
c) 240 units
d) 1,000 units

Break-even analysis

Use the passage and any figures or tables on the preceding page.

Question 3

What operating profit results from selling 900 units under the original conditions?

a) €18,000
b) €45,000
c) €33,000
d) €6,000

Question 4

If only the selling price rises to €55, what is the new break-even quantity?

a) 600 units
b) 800 units
c) 480 units
d) 218 units

Question 5

If only unit variable cost rises to €35, what is the new break-even quantity?

a) 600 units
b) 480 units
c) 343 units
d) 800 units

Question 6

At the original price and costs, capacity is limited to 500 units. Which conclusion follows?

a) Even at full capacity, the workshop makes a €2,000 operating loss.
b) Full capacity produces exactly zero operating profit.
c) Capacity makes fixed operating costs disappear.
d) Full capacity produces a €2,000 operating profit.

Accounting statements and transaction logic

A simplified balance sheet records assets, liabilities and equity at a point in time. The identity is assets = liabilities + equity. Initially, a business has €10,000 cash and €20,000 equipment, and owes €12,000. Assume these are its only assets and liabilities. Equity is the residual claim, not an additional asset.

An income statement records revenue and expenses over a period. Under the accrual approach used here, revenue is recognised when earned, even if cash will arrive later. The unpaid amount becomes a receivable, which is an asset. During a separate period, the business earns €8,000 revenue and incurs €5,000 expenses. Of the revenue, €5,000 is collected immediately and €3,000 remains receivable.

Borrowing creates cash and a liability; it is not sales revenue. Buying equipment for cash exchanges one asset for another and is not an immediate expense in this model. Depreciation is an expense that reduces the equipment’s carrying value without a cash payment at the moment it is recorded. Treat each question as a separate scenario, with no taxes, owner distributions or unmentioned transactions.

Question 1

What is initial equity?

a) €18,000
b) €42,000
c) €12,000
d) €30,000

Question 2

What is profit for the separate period described?

a) €0
b) €3,000
c) €5,000
d) €8,000

Accounting statements and transaction logic

Use the passage and any figures or tables on the preceding page.

Question 3

How much of the period’s revenue remains as a receivable?

a) €3,000
b) €0
c) €5,000
d) €8,000

Question 4

What is the immediate effect of receiving a new €4,000 loan in cash?

a) Revenue and profit each increase by €4,000.
b) Assets and liabilities each increase by €4,000; equity is unchanged.
c) Assets increase by €4,000 and liabilities decrease by €4,000.
d) Assets and equity each increase by €4,000; liabilities are unchanged.

Question 5

What is the immediate effect of buying €3,000 equipment for cash?

a) Total assets increase by €3,000.
b) Total equity falls by €3,000 because every cash payment is an expense.
c) Total assets remain unchanged; cash falls and equipment rises.
d) Liabilities necessarily increase by €3,000.

Question 6

What is the effect of recording an additional €1,000 depreciation expense?

a) Profit and equity decrease by €1,000, with no immediate cash outflow.
b) Cash and liabilities each decrease by €1,000.
c) Revenue increases by €1,000 while equity is unchanged.
d) Profit increases by €1,000 and cash decreases by €1,000.

Marketing mix, pricing and campaign comparisons

A retailer organises marketing decisions into four categories: product concerns the item’s features and quality; price concerns the amount charged; place concerns distribution and access; and promotion concerns communication intended to attract buyers. A single campaign may involve several categories, but a question may ask about one specific decision.

For a simplified campaign model, conversion rate is orders divided by visitors. Each order contains one unit. Campaign contribution after advertising equals (selling price - unit variable cost) × orders - advertising cost. Ignore fixed overhead, returns, taxes and repeat purchases. Unit variable cost is €8 in both campaigns. The table gives each campaign’s visitors, price, conversion rate and advertising cost.

Campaigns A and B ran in different weeks and reached different audiences. Their measured outcomes can be compared arithmetically, but a difference cannot automatically be attributed to price alone. A fair causal test of a price change would hold other relevant conditions comparable, for example through random allocation of similar visitors to the two offers.

Campaign	Visitors	Price	Conversion	Ad cost
A	1,000	€20	2%	€150
B	1,000	€16	3%	€150

Question 1

Which marketing-mix category primarily covers making the product more durable?

a) Price
b) Promotion
c) Product
d) Place

Question 2

Which category primarily covers adding a new retail distribution channel?

a) Price
b) Product
c) Promotion
d) Place

Marketing mix, pricing and campaign comparisons

Use the passage and any figures or tables on the preceding page.

Question 3

What is campaign A’s contribution after advertising under the stated model?

a) €150
b) €240
c) €90
d) €400

Question 4

How does campaign B’s contribution after advertising compare with A’s?

a) It is twice A’s.
b) It is €60 lower than A’s.
c) It is equal to A’s.
d) It is €60 higher than A’s.

Question 5

For campaign A’s listed orders and unit margin, what is the highest advertising cost that leaves a nonnegative campaign contribution?

a) €150
b) €240
c) €400
d) €90

Question 6

Which inference about the higher conversion rate in B is best supported?

a) Different audiences remove the need for a controlled comparison.
b) The comparison alone cannot isolate the effect of its lower price.
c) The lower price must be the sole cause of the higher conversion rate.
d) A higher conversion rate proves a higher campaign contribution.

Project scheduling and the critical path

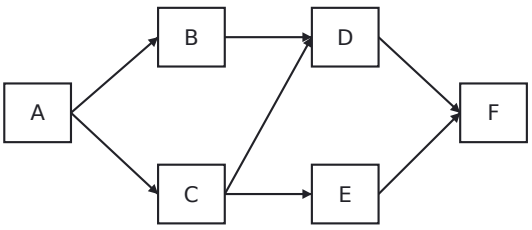
A project consists of activities A to F with the durations and dependencies in the table. Time is measured in days from project start, designated time 0. An activity can start as soon as all its predecessors finish. Activities on different branches may run simultaneously, and there are no resource limits. Durations are deterministic.

An activity’s earliest finish is its earliest start plus duration. If it has several predecessors, its earliest start is the latest of their finish times. The project ends when final activity F finishes. A path is a chain of dependent activities from start to finish; the longest path determines minimum project duration and is called a critical path.

An activity can sometimes be delayed without delaying the whole project because another branch is controlling the finish time. Shortening a noncritical activity may therefore leave the project duration unchanged. Each question about a changed duration is a separate scenario using the original table for every other activity. Assume activities can start immediately at a predecessor’s finish time, with no overnight waiting rule.

Activity	Duration	Predecessors
A	3	None
B	4	A
C	2	A
D	5	B and C
E	2	C
F	1	D and E

Dependencies (durations in table)



Question 1

What is the earliest completion time of the original project?

a) Day 10
b) Day 8
c) Day 17
d) Day 13

Question 2

Which is the original critical path?

a) A-B-D-F
b) A-B-E-F
c) A-C-E-F
d) A-C-D-F

Project scheduling and the critical path

Use the passage and any figures or tables on the preceding page.

Question 3

If only C takes one additional day, what happens to project duration?

a) It remains 13 days.
b) It becomes 15 days.
c) It becomes 12 days.
d) It becomes 14 days.

Question 4

If only B takes two additional days, when does the project finish?

a) Day 15
b) Day 13
c) Day 14
d) Day 17

Question 5

At what time can D start in the original schedule?

a) Day 3
b) Day 12
c) Day 7
d) Day 5

Question 6

If only E is shortened from two days to one day, what is the project duration?

a) 11 days
b) 14 days
c) 12 days
d) 13 days

Process capacity, bottlenecks and flow

A product passes through three stages in order: preparation, assembly and inspection. Their maximum sustained processing capacities are 60, 40 and 50 units per hour respectively. Every finished unit must pass through all three stages, with no rework or losses. Assume sufficient staffing within those stated capacities, no failures, and buffers that do not impose additional capacity limits.

The bottleneck is the stage with the lowest capacity. It limits the line’s maximum sustained output, even if other stages can work faster. Actual sustained output is also limited by demand: it cannot exceed the smaller of demand and line capacity. Stage utilisation is actual flow through the stage divided by that stage’s capacity.

For a stable process with consistent averaging boundaries, Little’s law relates average work in process L, throughput rate λ and average time in the system W: $L = \lambda W$. Time units must agree. Adding unfinished stock before a bottleneck does not itself increase that stage’s processing capacity. In improvement questions, change only the stage or quantity specified.

Question 1

What is the original line’s maximum sustained output?

a) 150 units/hour

b) 50 units/hour

c) 60 units/hour

d) 40 units/hour

Question 2

If assembly capacity rises to 55 units/hour, what is the new line capacity?

a) 55 units/hour

b) 50 units/hour

c) 40 units/hour

d) 60 units/hour

Process capacity, bottlenecks and flow

Use the passage and any figures or tables on the preceding page.

Question 3

Under the original capacities, what sustained output is possible when demand is only 35 units/hour?

a) 35 units/hour
b) 40 units/hour
c) 50 units/hour
d) 60 units/hour

Question 4

At that 35-unit/hour flow, what is assembly utilisation?

a) 87.5%
b) 114.3%
c) 70%
d) 100%

Question 5

A stable line has throughput 40 units/hour and average time in the system 30 minutes. What is average work in process?

a) 20 units
b) 40 units
c) 1,200 units
d) 80 units

Question 6

Which action alone cannot raise the original line’s maximum processing capacity?

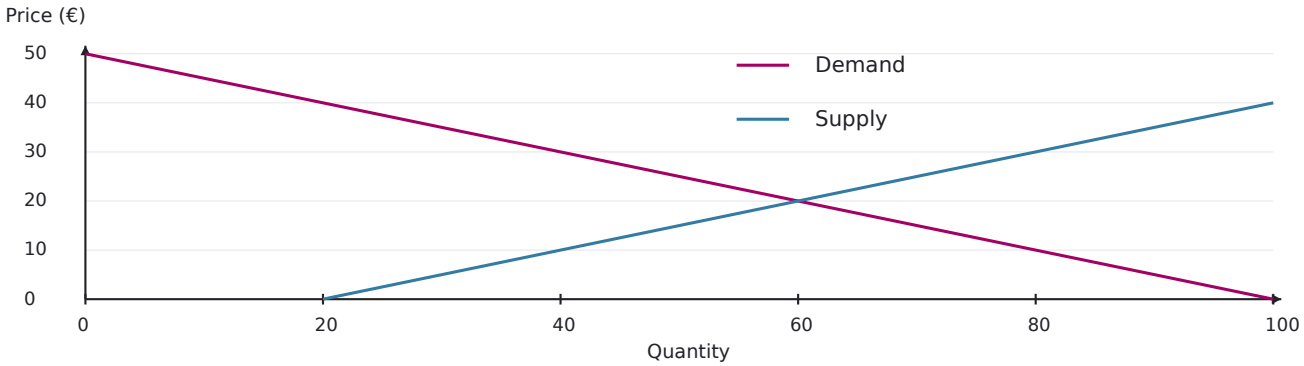
a) Increase assembly capacity above 50 while retaining the other capacities.
b) Increase assembly capacity from 40 to 45 units/hour.
c) Increase assembly capacity from 40 to 50 units/hour.
d) Place more waiting stock before assembly without changing any stage.

Supply, demand and market equilibrium

In a simplified market, quantity demanded is Qd = 100 - 2P and quantity supplied is Qs = 20 + 2P, where P is price in euros and quantities are units per period. Use only prices for which the relevant quantities are nonnegative. Market equilibrium occurs where Qd = Qs. A shortage is Qd - Qs when this difference is positive; a surplus is Qs - Qd when that difference is positive.

A change in the product's own price moves the market along a given demand or supply curve. A change in another determinant, such as buyers' preferences or production conditions, may shift a curve and therefore change its equation. Each shift question below starts from the original market and changes only the named curve.

A binding price ceiling fixes the highest permitted price below the original equilibrium in this model. When asked about the resulting shortage, calculate demand and supply at the ceiling rather than solving for a new unrestricted equilibrium. Ignore informal trading, rationing details and subsequent curve changes. The diagram shows the original inverse demand and supply relationships.



Question 1

What is the original equilibrium price?

a) €40
b) €30
c) €10
d) €20

Question 2

What is the original equilibrium quantity?

a) 80 units
b) 20 units
c) 40 units
d) 60 units

Supply, demand and market equilibrium

Use the passage and any figures or tables on the preceding page.

Question 3

What occurs at a price of €15 under the original curves?

a) A surplus of 20 units
b) A shortage of 20 units
c) A shortage of 50 units
d) A surplus of 70 units

Question 4

Demand changes to Qd = 120 - 2P while supply is unchanged. What is the new equilibrium price?

a) €30
b) €15
c) €25
d) €20

Question 5

Starting from the original market, supply changes to Qs = 40 + 2P. What is the new equilibrium quantity?

a) 40 units
b) 80 units
c) 60 units
d) 70 units

Question 6

A ceiling fixes the original market price at €10. What shortage follows?

a) 0 units
b) 20 units
c) 40 units
d) 60 units

Price elasticity and revenue

Price elasticity of demand measures the responsiveness of quantity demanded to price. Use the absolute magnitude of elasticity in this exercise. The midpoint formula is E = |(Q2 - Q1)/((Q1 + Q2)/2)| divided by |(P2 - P1)/((P1 + P2)/2)|. Demand is called elastic when E > 1, inelastic when E < 1, and unit elastic when E = 1.

A seller observes a move along an unchanged demand relationship from P1 = €10 and Q1 = 100 units to P2 = €12 and Q2 = 80 units. Revenue is price multiplied by quantity; it is not profit because costs have not been deducted.

For a small percentage price change, elasticity can also be used locally: percentage quantity change is approximately minus E times percentage price change. This approximation is distinct from the midpoint calculation for two observations. Perfectly inelastic demand has unchanged quantity as price changes and is drawn vertically when quantity is horizontal and price vertical. Keep other demand determinants fixed throughout the numerical scenarios.

Question 1

What is midpoint elasticity for the two observations, approximately?

a) 0.20
b) 2.00
c) 0.82
d) 1.22

Question 2

How is this demand response classified?

a) Inelastic
b) Elastic
c) Perfectly inelastic
d) Unit elastic

Price elasticity and revenue

Use the passage and any figures or tables on the preceding page.

Question 3

How does revenue change between the two observations?

a) It remains unchanged.
b) It falls by €40.
c) It falls by €200.
d) It rises by €40.

Question 4

For a separate local elasticity E = 0.5, a small 10% price increase implies approximately what quantity change?

a) A 5% increase
b) A 10% decrease
c) A 5% decrease
d) A 20% decrease

Question 5

Which description matches perfectly inelastic demand?

a) Quantity doubles whenever price doubles.
b) Quantity stays fixed as price changes.
c) Price stays fixed as quantity changes.
d) Quantity changes proportionally with price in the same direction.

Question 6

If local demand is unit elastic, what is the approximate first-order revenue effect of a small price increase?

a) Revenue rises by twice the price-change percentage.
b) Quantity remains exactly fixed.
c) Revenue necessarily becomes zero.
d) Revenue is approximately unchanged.

Cost curves and profit maximisation

A price-taking firm can sell any nonnegative continuous quantity q at a fixed market price of €12 per unit. Its total cost is TC(q) = 20 + 2q + q². The €20 term is fixed cost, incurred even when output is zero. Variable cost is VC(q) = 2q + q². Revenue is TR(q) = 12q and profit is TR - TC.

Marginal cost is the rate at which total cost increases with quantity; here MC(q) = 2 + 2q. Marginal revenue is MR = 12 because the selling price is constant. With this increasing marginal cost, an interior profit maximum occurs at MC = MR. For q > 0, average variable cost is VC/q and average total cost is TC/q.

A fixed-cost change alters total profit but not marginal cost. In a separate changed-price question, choose the best q ≥ 0 under the same cost equation. The continuous-quantity assumption avoids rounding issues. The model does not claim that marginal analysis guarantees positive profit: a best available output can still leave a loss after fixed costs.

Question 1

What output maximises profit at the original price?

a) 6 units
b) 4 units
c) 10 units
d) 5 units

Question 2

What is total cost at q = 5?

a) €45
b) €55
c) €35
d) €60

Cost curves and profit maximisation

Use the passage and any figures or tables on the preceding page.

Question 3

What is profit at q = 5?

a) -€5
b) €5
c) €55
d) €25

Question 4

If only fixed cost rises from €20 to €40, what happens to the profit-maximising output?

a) It remains 5 units.
b) It falls to 2.5 units.
c) It rises to 10 units.
d) It falls to 0 units automatically.

Question 5

What is average variable cost at q = 5?

a) €12 per unit
b) €5 per unit
c) €11 per unit
d) €7 per unit

Question 6

If the selling price falls to €2 with the original cost equation retained, which nonnegative output maximises profit?

a) 1 unit
b) 10 units
c) 5 units
d) 0 units

GDP, inflation and unemployment measures

A simplified economy produces one final good. Nominal GDP values current production at current prices. Real GDP values production at a fixed base-year price. Year 1 is the base year: it produces 100 units at €10 each. Year 2 produces 110 units at €12 each. For this one-good illustration, the price index in year 1 is 100 and in year 2 is (current price/base price) × 100. Inflation is the percentage increase in the price index, not the percentage increase in nominal GDP.

In a separate labour-market snapshot, 920 people are employed, 80 have no job but are actively seeking work, and 500 are outside the labour force. The labour force includes employed people and unemployed active jobseekers. The unemployment rate is unemployed active jobseekers divided by the labour force.

These definitions distinguish current production from cash transfers. A transfer payment without a newly produced good or service is not itself part of GDP. Likewise, a person leaving active job search changes the unemployment-rate denominator as well as its numerator. Treat each labour-market change as a separate scenario.

Year	Quantity	Price
1	100	€10
2	110	€12

Question 1

What is nominal GDP in year 2?

a) €1,100
b) €1,000
c) €1,200
d) €1,320

Question 2

What is real GDP in year 2 using year-1 prices?

a) €1,000
b) €1,100
c) €1,320
d) €1,200

GDP, inflation and unemployment measures

Use the passage and any figures or tables on the preceding page.

Question 3

What is real GDP growth from year 1 to year 2?

a) 12%
b) 32%
c) 20%
d) 10%

Question 4

What is inflation in this one-good illustration?

a) 10%
b) 2%
c) 20%
d) 32%

Question 5

What is the unemployment rate in the original snapshot?

a) 8.70%
b) 16%
c) 5.33%
d) 8%

Question 6

Twenty of the 80 jobseekers stop seeking work, with no employment change. Which statement is correct?

a) The unemployment rate remains 8% because employment is unchanged.
b) The unemployment rate falls even though no additional person becomes employed.
c) The number employed increases to 940.
d) The labour force increases to 1,020.

Market structures and monopoly decisions

In the simplified market classifications used here, perfect competition has many sellers of identical products, and each individual firm takes the market price as given. A monopoly has a single seller protected by an entry barrier, such as exclusive control of a necessary input. Oligopoly has a small number of important sellers whose decisions can depend on anticipated rival responses. Monopolistic competition combines many sellers with differentiated products.

For one hypothetical monopoly, inverse demand is $P = 20 - q$, total revenue is $TR = 20q - q^2$, marginal revenue is $MR = 20 - 2q$, and marginal cost is constant at $MC = 4$. The firm chooses a continuous $q \geq 0$ and charges the price on its demand curve. The interior profit-maximising condition is $MR = MC$; once q is found, use the demand equation to find price.

Market power does not guarantee positive profit because fixed costs can still exceed operating surplus. As a comparison benchmark, suppose a competitive industry with the same market demand has a horizontal supply at $P = MC = 4$. Ignore capacity limits, taxes and any unmentioned changes.

Question 1

Which market description makes an individual seller a price taker in this passage?

a) Monopoly
b) Monopolistic competition in every case
c) Oligopoly in every case
d) Perfect competition

Question 2

For the monopoly at any positive q, how does marginal revenue compare with price?

a) Marginal revenue equals price.
b) Marginal revenue is twice price.
c) Marginal revenue is below price.
d) Marginal revenue is above price.

Market structures and monopoly decisions

Use the passage and any figures or tables on the preceding page.

Question 3

What quantity maximises the monopoly’s profit under the stated model?

a) 10 units
b) 8 units
c) 16 units
d) 4 units

Question 4

What price does the monopoly then charge?

a) €8
b) €16
c) €4
d) €12

Question 5

What quantity is demanded under the competitive benchmark price of €4?

a) 8 units
b) 12 units
c) 16 units
d) 4 units

Question 6

A market has a few major firms that anticipate one another’s price responses. Which classification best fits?

a) Monopoly
b) Perfect competition
c) A market with no strategic interaction
d) Oligopoly

Game theory and payoff matrices

Two organisations simultaneously choose whether to cooperate (C) or defect (D) in a one-round project. Neither sees the other's current choice before acting. Each cell in the table lists (payoff to Row, payoff to Column), and each organisation prefers a larger own payoff. There is no binding agreement and no future round.

A best response is a choice that gives the highest own payoff given the other organisation's choice. A strictly dominant strategy gives a higher payoff regardless of what the other chooses. A Nash equilibrium is a pair of choices from which neither organisation can increase its own payoff by changing its choice alone while the other choice stays fixed.

Joint payoff is the sum of both listed payoffs, but each organisation's individual incentive is evaluated using only its own payoff. A jointly better result therefore need not be a Nash equilibrium. In the last question, change only the mutual-cooperation cell from (3,3) to (4,4), retaining all other cells and the one-round structure.

Row choice	Column C	Column D
C	(3, 3)	(0, 5)
D	(5, 0)	(1, 1)

Question 1

If Column chooses C, what is Row's best response?

a) Either choice, because both give Row 3
b) D, giving Row 5
c) C, giving Row 5
d) C, giving Row 3

Question 2

Which strategy is strictly dominant for Row in the original game?

a) D
b) C
c) Both C and D
d) Neither C nor D

Game theory and payoff matrices

Use the passage and any figures or tables on the preceding page.

Question 3

Which choice pair is the original Nash equilibrium?

a) (D, C)
b) (C, D)
c) (D, D)
d) (C, C)

Question 4

Which original outcome gives the greatest joint payoff?

a) (D, C)
b) (C, D)
c) (D, D)
d) (C, C)

Question 5

At (D, D), what happens to Row’s payoff if only Row switches to C?

a) It rises from 1 to 5.
b) It rises from 1 to 3.
c) It falls from 1 to 0.
d) It remains at 1.

Question 6

After the specified change to mutual cooperation, what happens to the Nash equilibrium?

a) (C, C) becomes the unique Nash equilibrium.
b) (D, D) remains the unique Nash equilibrium.
c) Only (C, D) becomes a Nash equilibrium.
d) Every cell becomes a Nash equilibrium.

Qualitative and quantitative research strategies

A research team investigates how a university mentoring programme relates to students' transition into employment. One strand starts with a pre-specified hypothesis and analyses standardised records for 1,500 students. This is a quantitative, deductive strand: it tests a proposed relationship using numerical data. Since participation was not randomly assigned, an observed association alone does not establish the programme's causal effect.

A second strand conducts detailed interviews with 12 students to explore how mentoring might influence confidence, information and job-search decisions. This qualitative, inductive strand can develop explanations of mechanisms and identify relevant conditions. Its deliberately selected cases do not by themselves establish how common those mechanisms are in the whole student population.

Both strands involve a research question, a design, data collection and analysis, and reporting. Qualitative inquiry may refine questions as new themes appear, provided changes are documented. Quantitative projects should distinguish pre-specified tests from exploratory analyses made after inspecting results; later changes are not automatically forbidden, but should be reported transparently. Combining the two strands is a mixed-methods design, intended to connect numerical patterns with accounts of how those patterns might arise.

Question 1

Which description best fits the strand using 1,500 standardised records?

a) An interview-only case study
b) Qualitative and purely inductive
c) A study with no pre-specified hypothesis
d) Quantitative and deductive

Question 2

What is the interview strand particularly suited to investigating?

a) How mentoring may influence students' decisions
b) A causal effect guaranteed by random assignment
c) The exact population prevalence of every mechanism
d) The national unemployment rate without further data

Qualitative and quantitative research strategies

Use the passage and any figures or tables on the preceding page.

Question 3

A new interview theme suggests refining the qualitative question. Which response is appropriate?

a) Treat the new theme as a proven population-wide mechanism.
b) Document the change and explain how it affects the inquiry.
c) Discard every interview solely because the question evolved.
d) Change the question silently so the report appears fully pre-planned.

Question 4

Which reporting practice most clearly misrepresents the quantitative strand?

a) Reporting an unsuccessful test of the original hypothesis
b) Documenting a justified change to a measurement procedure
c) Presenting a hypothesis invented after seeing the results as if it were pre-specified
d) Separating exploratory findings from the original test

Question 5

Which combination best supports both the relationship and mechanism questions?

a) Interview one administrator without examining student experiences.
b) Analyse standardised student records and conduct detailed participant interviews.
c) Count records twice using the same measure.
d) Use only a list of programme publicity slogans.

Question 6

Mentored students have higher employment rates in the records. What follows from this result alone?

a) The interview strand is no longer relevant.
b) There is an observed association, but other explanations remain possible.
c) Every interviewed mechanism must operate in all students.
d) Mentoring has been proved to cause the entire difference.

Sampling and statistical significance

A college population consists of 80% daytime students and 20% evening students. Researchers select students randomly within each group but deliberately obtain equal-sized samples from the two groups. This is stratified sampling with disproportionate allocation. In the collected data, 40% of daytime students and 80% of evening students support a proposal. To estimate population support, weight group estimates by population shares rather than giving both groups equal weight.

In a separate pre-specified statistical test, the null hypothesis states that two population rates are equal. The p-value is 0.03 and the significance threshold α is 0.05. A p-value is the probability, under the null model and its assumptions, of results at least as extreme as those observed. It is not the probability that the null hypothesis is true. The decision rule is to reject the null when $p < \alpha$.

A Type I error is rejecting a true null hypothesis. A larger well-designed sample can reduce random sampling error, but merely enlarging a systematically unrepresentative sample does not guarantee removal of selection bias. Treat sampling design and statistical significance as separate aspects of study quality.

Question 1

What sampling method describes random selection within daytime and evening groups?

a) A convenience sample with no random selection
b) Stratified sampling
c) Simple random sampling with no grouping
d) A census of every student

Question 2

What is the correctly weighted estimate of population support?

a) 80%
b) 60%
c) 48%
d) 40%

Sampling and statistical significance

Use the passage and any figures or tables on the preceding page.

Question 3

What does p = 0.03 mean under the definition supplied?

a) There is a 0.97 probability that the research hypothesis is true.
b) Under the null model, results at least as extreme have probability 0.03.
c) Exactly 3% of participants supplied incorrect answers.
d) There is a 0.03 probability that the null hypothesis is true.

Question 4

What decision follows from the stated threshold rule?

a) Reject the null hypothesis.
b) Conclude the null is certainly false.
c) Change α after viewing the result.
d) Retain the null because 0.03 exceeds 0.05.

Question 5

Which event is a Type I error?

a) Rejecting the null when the population rates are actually equal
b) Failing to reject the null when the rates differ
c) Reporting the p-value alongside the threshold
d) Drawing a larger random sample

Question 6

A researcher recruits only volunteers from one particularly enthusiastic club. What does increasing that sample’s size guarantee?

a) It guarantees a causal explanation.
b) It guarantees representativeness of the whole college.
c) It does not guarantee removal of the selection bias.
d) It makes population weighting unnecessary in every study.

Correlation, confounding and causal claims

A cross-sectional survey finds a positive correlation between library visits and examination scores. The correlation coefficient r measures the direction and strength of linear association and lies between -1 and 1. Its sign gives direction, but it is not a percentage of students benefiting and does not itself measure a causal effect. A zero correlation rules out linear association as measured by r, not every possible nonlinear relationship.

Motivation may influence both library attendance and examination performance. Such a common cause is a potential confounder. Reverse causation is another possibility: students' existing academic experiences may influence whether they visit the library. Cross-sectional measurements taken at one time may not establish which factor came first.

To investigate an intervention, a separate study randomly assigns comparable students either to receive a library-orientation offer or to receive no offer. Suppose assignment is implemented correctly, there is no differential loss to follow-up or interference between students, and outcomes are measured consistently. This design supports a causal comparison of the offer. It does not automatically identify the effect of each additional library visit, because visiting remains a separate behaviour.

Question 1

What does a positive r between visits and scores describe?

a) Every extra visit causes the same score increase.

b) Higher visits tend to accompany higher scores in a linear association.

c) All frequent visitors score above all infrequent visitors.

d) The proportion who benefit equals r.

Question 2

Why is motivation a potential confounder in the survey?

a) It is necessarily caused only by library visits.

b) It prevents either variable from being recorded.

c) It may influence both library visits and scores.

d) It is measured in different units from examination scores.

Correlation, confounding and causal claims

Use the passage and any figures or tables on the preceding page.

Question 3

Which example illustrates the reverse-causation concern?

a) Motivation independently influences attendance and scores.
b) Previous academic difficulties influence later library attendance.
c) An arithmetic error changes a recorded score.
d) A random assignment determines who receives the offer.

Question 4

Under the stated assumptions, what effect does the randomised study most directly identify?

a) The effect of each additional library visit in every student
b) The effect of receiving the orientation offer
c) The effect of motivation independent of all other factors
d) The effect of library construction on national scores

Question 5

Why should a one-time survey be cautious about causal direction?

a) It may not establish which variable changed first.
b) It makes every common cause impossible.
c) It guarantees that neither variable affects the other.
d) It cannot calculate any numerical association.

Question 6

If a different dataset has r = 0, which conclusion is justified?

a) Every recorded value must be identical.
b) It has no linear association as measured by r, but a nonlinear relationship may remain.
c) Its variables must be independent in every respect.
d) Changing one variable can never affect the other.

Survey and questionnaire design

A transport survey aims to measure passengers' experiences consistently. A question should target one construct at a time. Asking whether a service is "reliable and affordable" combines two judgements and may be difficult to answer when a passenger agrees with only one. Leading wording can encourage a preferred response; neutral wording avoids embedding an evaluation such as "excellent" in the question itself.

Response categories should be mutually exclusive and collectively cover the intended responses. A specified recall period helps respondents interpret frequency consistently. For instance, a question about trips in the last seven days is more precise than one asking about trips "recently". Nonresponse and socially desirable reporting can still affect results even when questions are carefully written.

To compare survey waves, keep measurement wording and procedures consistent or document changes that may affect comparability. A pilot study can reveal ambiguous wording before full collection. Randomly assigning respondents from the same target population to alternative wordings, with otherwise identical procedures, helps assess wording effects. Anonymity may reduce fear of judgement but does not guarantee honest or complete responses.

Question 1

What is the main problem with "Was the service reliable and affordable?"

a) It specifies too precise a time period.

b) It uses response categories that are necessarily overlapping.

c) It asks about two judgements in one item.

d) It is an experiment rather than a survey question.

Question 2

Which wording is the most neutral assessment of punctuality?

a) Why was the service so disappointingly late?

b) How much did you appreciate the excellent punctuality?

c) How would you rate the service's punctuality?

d) Do you agree that the punctuality was outstanding?

Survey and questionnaire design

Use the passage and any figures or tables on the preceding page.

Question 3

Which categories avoid overlap and cover every nonnegative whole-number trip count?

a) 0; 1; 2; 3-4
b) 0-2; 2-4; 4 or more
c) 0; 1-2; 3-4; 5 or more
d) 1-2; 3-4; 5 or more

Question 4

Which revision most directly standardises the recall period?

a) Ask about trips during the last seven days.
b) Remove all references to time.
c) Ask whether passengers are “regular users”.
d) Ask whether passengers travel “quite often”.

Question 5

Why may changing wording between survey waves complicate trend interpretation?

a) A response change may reflect wording rather than a change in experience.
b) Wording changes automatically eliminate sampling error.
c) Changing wording proves that respondents were previously dishonest.
d) Two waves can never be compared for any purpose.

Question 6

Which design best isolates an effect of question wording?

a) Use one wording this year and another in a different city next year.
b) Change the wording, sample and response scale simultaneously.
c) Give the preferred wording only to satisfied passengers.
d) Randomly assign similar respondents to two wordings with otherwise identical procedures.

Argument analysis and logical validity

An argument contains premises and a conclusion. A deductively valid argument is one in which the conclusion cannot be false if all premises are true. Validity concerns the reasoning structure; it does not establish that the premises are factually true. For a conditional statement “if P, then Q”, P is sufficient for Q and Q is necessary for P. The statement alone does not establish “if Q, then P”.

Two valid patterns are: if P then Q; P; therefore Q, and if P then Q; not Q; therefore not P. Inferring P merely from Q is invalid because another situation may also produce Q.

Suppose every approved design has passed a load test. Passing is therefore necessary for approval, but the passage does not say it is sufficient. A design may pass and still fail some other approval requirement. In evaluating non-deductive arguments, focus on relevant evidence. An ad hominem attack dismisses a claim by attacking the person instead of addressing the reasoning. A straw-man response attacks a distorted version of the original claim.

Question 1

A design is approved. What follows from the stated premise?

a) It cannot have any design limitation.

b) It has passed the load test.

c) It is the only design that passed.

d) It passed every conceivable test.

Question 2

Design K passed the load test. What can be concluded about its approval from that fact alone?

a) Its approval is not determined.

b) It must be approved.

c) It must be rejected.

d) No other design can be approved.

Argument analysis and logical validity

Use the passage and any figures or tables on the preceding page.

Question 3

Design L did not pass the load test. What follows?

a) L is approved.
b) Every other design is approved.
c) The conditional premise says nothing about L.
d) L is not approved.

Question 4

Which possible case disproves the claim that passing alone guarantees approval?

a) A design with no recorded test result
b) A design that failed and was not approved
c) A design that passed and was approved
d) A design that passed but was not approved

Question 5

A reviewer rejects a cost estimate solely because its author is personally unlikeable. What is the reasoning error?

a) A direct numerical comparison
b) A valid contrapositive
c) Ad hominem
d) A necessary-condition inference

Question 6

A proposal to reduce unnecessary printing is criticised as a proposal to ban all communication. What error is illustrated?

a) An accurate restatement of the proposal
b) A straw-man response
c) An ad hominem attack
d) A valid contrapositive

Social and psychological concepts in applied research

A campus study investigates reusable-cup returns. A descriptive social norm communicates what other people commonly do, while an injunctive norm communicates what people approve of or expect. The intervention displays the truthful message “Most students return their reusable cups”. Researchers randomly assign 500 students to a setting showing this message and 500 to an otherwise comparable setting with no message. Assume assignment is implemented properly, all return outcomes are recorded, and there is no communication between groups during the study.

The table reports observed returns. Counting actual returns measures behaviour; asking whether returning cups is important measures an attitude. Self-reports can be affected by social desirability, where respondents present themselves favourably. This possibility does not prove that every disagreement between a report and a count is deliberate dishonesty.

Random assignment strengthens a causal comparison of the message in these settings, but the result alone does not identify the exact psychological mechanism or guarantee the same effect on every campus. A reward contingent on returning a cup is a different intervention: if it increases future returning, it functions as positive reinforcement. No reward is included in the reported experiment.

Condition	Students	Cups returned
Message	500	300
No message	500	200

Question 1

What kind of norm does “Most students return their reusable cups” communicate?

a) A monetary reward
b) A descriptive norm
c) A statement of individual ability
d) An injunctive norm only

Question 2

What is the observed return rate in the message group?

a) 20%
b) 50%
c) 40%
d) 60%

Social and psychological concepts in applied research

Use the passage and any figures or tables on the preceding page.

Question 3

What is the difference in return rates, in percentage points?

a) 100 percentage points
b) 20 percentage points
c) 10 percentage points
d) 50 percentage points

Question 4

Which measurement most directly captures the target behaviour?

a) A rating of the message’s appearance
b) Agreement that environmental care is important
c) Whether students recognise the campus logo
d) The number of cups actually returned

Question 5

Which conclusion is best supported by this experiment under the stated assumptions?

a) The message works identically for every student on every campus.
b) The experiment establishes the exact psychological process in each participant.
c) The results support an effect of the message here; wider effects and mechanisms need further evidence.
d) The experiment proves that a monetary reward caused the difference.

Question 6

In a separate intervention, a small reward is given after a cup return and future returning increases. Which concept fits?

a) Random sampling
b) Positive reinforcement
c) An absence of behavioural measurement
d) A descriptive norm with no consequence

Answers by exercise and question

Exercise / topic	1	2	3	4	5	6
01 Vector products, angles, areas and volumes	a	c	d	c	a	c
02 Systems of linear equations and algebra	a	a	b	b	c	c
03 Functions and graphs	b	c	c	c	b	b
04 Differentiation and integration	d	c	a	a	b	c
05 Probability and descriptive statistics	c	c	c	b	d	c
06 Matrices and determinants	d	a	a	d	c	a
07 Kinematics and Newton’s laws	a	d	d	a	b	c
08 Hydrostatics, pressure and buoyancy	c	c	c	b	a	c
09 Heat, temperature and the ideal gas law	d	b	a	c	d	d
10 Electrical quantities, Ohm’s law and power	b	c	b	d	c	d
11 Stoichiometry, concentration and pH	a	d	c	c	d	d
12 Waves and mechanical energy	b	a	a	d	a	b
13 Statics: forces, moments and equilibrium	b	a	c	b	a	d
14 Stress, strain and basic beam behaviour	a	a	a	a	d	d
15 Series, parallel and Kirchhoff’s laws	a	b	a	a	c	a
16 Energy conversion and efficiency	d	a	c	b	d	d
17 Control and feedback loops	a	b	c	c	b	a
18 Unit conversions and dimensional analysis	a	c	b	a	d	a
19 Binary and hexadecimal number systems	b	d	d	d	a	b
20 Boolean logic and logic gates	c	b	a	a	b	a
21 Algorithms and step-by-step tracing	b	d	b	a	a	d
22 Algorithmic complexity and growth	b	d	c	b	d	a
23 Arrays, stacks and queues	b	d	a	c	d	c
24 Recursion and iteration	d	a	c	c	b	c
25 Optimal order quantity and inventory costs	d	b	c	a	a	c
26 Break-even analysis	d	b	d	c	d	a
27 Accounting statements and transaction logic	a	b	a	b	c	a
28 Marketing mix, pricing and campaign comparisons	c	d	c	c	b	b
29 Project scheduling and the critical path	d	a	a	a	c	d
30 Process capacity, bottlenecks and flow	d	b	a	a	a	d
31 Supply, demand and market equilibrium	d	d	b	c	d	c
32 Price elasticity and revenue	d	b	b	c	b	d
33 Cost curves and profit maximisation	d	b	b	a	d	d
34 GDP, inflation and unemployment measures	d	b	d	c	d	b
35 Market structures and monopoly decisions	d	c	b	d	c	d
36 Game theory and payoff matrices	b	a	c	d	c	b
37 Qualitative and quantitative research strategies	d	a	b	c	b	b
38 Sampling and statistical significance	b	c	b	a	a	c
39 Correlation, confounding and causal claims	b	c	b	b	a	b
40 Survey and questionnaire design	c	c	c	a	a	d
41 Argument analysis and logical validity	b	a	d	d	c	b
42 Social and psychological concepts in applied research	b	d	b	d	c	b

Exercise 1 · Vector products, angles, areas and volumes

1. a) Multiply corresponding components and add: $2 \times 1 + 0 \times 3 + 0 \times 0 = 2$.

2. c) The lengths are 2 and $\sqrt{10}$. Thus $\cos \theta = 2/(2\sqrt{10}) = 1/\sqrt{10}$.

3. d) The first two cross-product components are zero. The third is $2 \times 3 - 0 \times 1 = 6$.

4. c) The area is the magnitude of (0, 0, 6), which is 6. Half of this would be the triangle area.

5. a) The base parallelogram has area 6 and c is perpendicular to it with length 4. Equivalently, the absolute triple product is 24.

6. c) All three vectors now have zero z-component. They lie in the xy-plane, so their scalar triple product and enclosed volume are zero.

Exercise 2 · Systems of linear equations and algebra

1. a) Subtract $x + y = 14$ from $2x + y = 20$. The y terms cancel and $x = 6$.

2. a) The original equations give $x = 6$ and $y = 8$. Therefore $z = 3 \times 6 - 8 = 10$.

3. b) Both original equations have coefficient 1 on y. Subtracting the complete first equation cancels y without changing the solution set.

4. b) Subtracting the retained first equation gives the new $x = 22 - 14 = 8$. The original x was 6, so the increase is 2.

5. c) Multiplication by 2 must apply to every term on both sides. It gives $2x + 2y = 28$.

6. c) Doubling the first equation gives $2x + 2y = 28$, which contradicts 30. No pair can satisfy both constraints.

Exercise 3 · Functions and graphs

1. b) The linear function $f(x) = 2x + 1$ increases by 2 for every unit of x. The other functions do not have a constant slope.

2. c) The squared term is nonnegative and becomes zero at $x = 2$. This is the vertex and minimum.

3. c) The exponential rule gives $h(3) = 2^3 = 8$.

4. c) Since $2^3 = 8$, $\log_2(8) = 3$.

5. b) Adding 3 to the entire output gives $f(x) + 3 = 2x + 4$. Changing the input or slope would describe a different transformation.

6. b) At $x = 0$, both outputs are 1. The other listed pairs are (3,2), (5,4) and (7,8), none of which are equal.

Exercise 4 · Differentiation and integration

1. d) Differentiate to obtain $v(t) = 2t + 2$. At $t = 2$, this is 6 m/s.

2. c) The initial position is 5 and the final position is 20. Their difference, or the integral of velocity, is 15 m.

3. a) Average velocity equals displacement divided by time: $15/3 = 5$ m/s. The initial position is not part of the displacement.

4. a) Changing the constant shifts the position reference. Its derivative is zero, so $v(t)$ remains $2t + 2$.

5. b) Velocity stays positive on $0 \le t \le 3$. Therefore its integral gives both displacement and distance, unlike an integral of position.

6. c) Differentiate $v(t) = 2t + 2$ once more. The acceleration is 2 m/s² throughout the interval.

Exercise 5 · Probability and descriptive statistics

1. c) The total is 2 + 4 + 4 + 6 = 16. Dividing by four outcomes gives a mean of 4.

2. c) The squared deviations from 4 are 4, 0, 0 and 4. Their mean is 2, so the standard deviation is √2.

3. c) The interval is 10 ± 2, exactly one standard deviation on either side of the mean. The passage assigns approximately 68% to this interval.

4. b) Multiply the independent probabilities: 0.2 × 0.2 = 0.04.

5. d) Every reading and the mean increase by 3. Each reading therefore has the same deviation as before, leaving the variance unchanged.

6. c) Three of the four listed outcomes qualify: 4, 4 and 6. Repeated values still represent separate equally likely outcomes.

Exercise 6 · Matrices and determinants

1. d) The two rows give 2 × 3 + 2 = 8 and 3 + 2 = 5.

2. a) For M, ad - bc = 2 × 1 - 1 × 1 = 1.

3. a) The determinant is 1. Swap the diagonal entries and negate the off-diagonal entries to obtain [[1,-1],[-1,2]].

4. d) Its determinant is 2 × 2 - 4 × 1 = 0. This establishes noninvertibility but does not mean every output is zero.

5. c) Doubling one row doubles the determinant. The original determinant is 1, so the new value is 2.

6. a) Multiplication on the right by N scales the second column of M. Multiplication on the left scales its second row. Thus the order matters.

Exercise 7 · Kinematics and Newton’s laws

1. a) The net horizontal force is 8 - 2 = 6 N. Dividing by 2 kg gives 3 m/s².

2. d) Starting from rest, v = 0 + 3 × 4 = 12 m/s.

3. d) Use s = ½at² because u = 0: ½ × 3 × 4² = 24 m.

4. a) A third-law pair acts on two different objects. Only forces acting on the trolley enter its own net-force calculation.

5. b) Removing both horizontal forces makes the net force zero. Acceleration is then zero, but the velocity already acquired remains.

6. c) The net force remains 6 N. With double the mass, acceleration becomes 6/4 = 1.5 m/s².

Exercise 8 · Hydrostatics, pressure and buoyancy

1. c) Add surface pressure to the water contribution: 100,000 + 1,000 × 10 × 5 = 150,000 Pa.

2. c) Gauge pressure excludes surface pressure. Thus it is ρgh = 50,000 Pa.

3. c) Floating requires Vsub = m/ρ = 12/1,000 = 0.012 m³. Divide by total volume 0.020 m³ to obtain 0.60.

4. b) Full submersion displaces 0.020 m³, giving ρgV = 1,000 × 10 × 0.020 = 200 N. This is greater than the block’s weight.

5. a) Constant pV gives V2 = p1V1/p2 = 1 × 2/2 = 1 L. Both pressures are absolute.

6. c) Equilibrium requires Vsub = 12/800 = 0.015 m³. This is below the block’s total volume, so floating is possible.

Exercise 9 · Heat, temperature and the ideal gas law

1. d) Rearrange $Q = mc\Delta T$: $\Delta T = 80{,}000/(2 \times 4{,}000) = 10$ K.

2. b) At fixed V and n, pressure is proportional to absolute temperature. The temperature ratio is 600/300 = 2.

3. a) With the stated sign convention, $\Delta U = Q - W = 500 - 200 = 300$ J.

4. c) Insulation gives Q = 0, not ΔU = 0. Hence $\Delta U = 0 - 100 = -100$ J.

5. d) Convert the Celsius temperature to an absolute temperature: 27 + 273 = 300 K.

6. d) At fixed pressure and gas amount, V/T is constant. Thus V2 = 2 × 450/300 = 3 L.

Exercise 10 · Electrical quantities, Ohm’s law and power

1. b) Ohm’s law gives $I = V/R = 12/6 = 2$ A.

2. c) Use $P = V^2/R = 12^2/6 = 24$ W, equivalently 12 V × 2 A.

3. b) At a constant 24 W, $E = Pt = 24 \times 5 = 120$ J.

4. d) With R fixed, power is proportional to V^2. Doubling V multiplies power by $2^2 = 4$.

5. c) The total series resistance is 2 + 4 = 6 Ω. The circuit current is 12/6 = 2 A and passes through both resistors.

6. d) The energy is 24 W × 2 h = 48 Wh. Divide by 1,000 to obtain 0.048 kWh.

Exercise 11 · Stoichiometry, concentration and pH

1. a) The 2 mol O_2 consume only 4 mol H_2. Hydrogen is available in excess, so oxygen is exhausted first.

2. d) The ratio is 2 mol water per 1 mol oxygen. The limiting 2 mol oxygen therefore produce 4 mol water.

3. c) Use c = n/V = 0.50/2.0 = 0.25 mol/L.

4. c) The solute amount is 0.50 × 0.20 = 0.10 mol. Dividing by the final 1.0 L gives 0.10 mol/L.

5. d) The base-10 logarithm is -3, so its negative is 3.

6. d) The concentration increases tenfold. Applying the logarithmic definition gives pH 2 instead of 3.

Exercise 12 · Waves and mechanical energy

1. b) Multiply frequency by wavelength: v = 5 × 2 = 10 m/s.

2. a) Ek = ½ × 2 × 3^2 = 9 J. Speed must be squared.

3. a) Ep = mgh = 2 × 10 × 5 = 100 J.

4. d) Set mgh = ½mv^2. Mass cancels, giving v = √(2gh) = √100 = 10 m/s.

5. a) The stated wave model makes energy proportional to amplitude squared. Doubling amplitude therefore gives a factor of four.

6. b) Rearrange v = fλ as λ = v/f. With v fixed, doubling f halves λ.

Exercise 13 · Statics: forces, moments and equilibrium

1. b) Vertical equilibrium requires the upward support reactions to balance both downward loads: 60 + 40 = 100 N.

2. a) Moments about L give RR = (60 × 1 + 40 × 3)/4 = 45 N.

3. c) The added moment about L is 60 × 1 = 60 N·m. Divide by the 4 m span to obtain a 15 N increase in RR.

4. b) Moment equals force times perpendicular lever arm: 10 × 2 = 20 N·m.

5. a) A couple can have zero net force but a nonzero moment. Both force and moment balance are required for static equilibrium.

6. d) Use the magnitude of one force times the separation: 20 × 0.5 = 10 N·m. Adding the force magnitudes would double-count the couple.

Exercise 14 · Stress, strain and basic beam behaviour

1. a) Using N and mm^2 gives σ = 2,000/20 = 100 N/mm^2 = 100 MPa.

2. a) Use the same length units: ε = 0.50/1,000 = 0.0005.

3. a) E = σ/ε = 100/0.0005 = 200,000 MPa, or 200 GPa.

4. a) Doubling area halves stress. With E fixed, strain also halves; since original length is fixed, elongation halves.

5. d) Deflection is proportional to L^3. Doubling length gives a factor of $2^3 = 8$.

6. d) The passage explicitly limits this relation to linear elastic behaviour. Units can be converted, but that cannot restore a failed physical assumption.

Exercise 15 · Series, parallel and Kirchhoff’s laws

1. a) $1/R_p = 1/6 + 1/3 = 1/2$, so $R_p = 2\ \Omega$.

2. b) Total resistance is $R1 + R_p = 2 + 2 = 4\ \Omega$. The source current is $12/4 = 3$ A.

3. a) R1 drops $3\ \text{A} \times 2\ \Omega = 6$ V. The remaining 6 V is across both parallel branches.

4. a) The branch voltage is 6 V. Thus $I3 = 6/3 = 2$ A. The other branch carries 1 A.

5. c) R1 and R2 remain in series with total resistance 8 Ω. Current is $12/8 = 1.5$ A.

6. a) The 3 A entering the junction divides into branch currents of 1 A and 2 A. The voltage equation instead expresses loop voltage balance.

Exercise 16 · Energy conversion and efficiency

1. d) Motor efficiency is $800/1{,}000 = 0.80$, or 80%.

2. a) The transmission receives 800 J and delivers 600 J. Its efficiency is $600/800 = 75\%$.

3. c) Either divide 600 by 1,000 or multiply 0.80 by 0.75. Both give 60%. Efficiencies of sequential stages are not added.

4. b) The total loss from the useful output is $1{,}000 - 600 = 400$ J, including losses from both stages.

5. d) The motor still delivers 800 J. The improved transmission delivers $0.90 \times 800 = 720$ J.

6. d) For steady power, efficiency is useful output divided by input: $300/500 = 60\%$.

Exercise 17 · Control and feedback loops

1. a) Subtract the measurement from the target: e = 60 - 52 = 8 °C.

2. b) The computed command is 2 × 8 = 16, which lies within the actuator limits and is not clipped.

3. c) The new error is 3 °C, so the command is 2 × 3 = 6 units.

4. c) The report equals actual temperature plus 5. Therefore actual temperature is 60 - 5 = 55 °C. This does not assume steady-state error is zero.

5. b) Open-loop operation omits measurement-based adjustment. Saturation or a different gain does not itself remove the feedback loop.

6. a) The computed command is 30, but the actuator clips it at the maximum of 20.

Exercise 18 · Unit conversions and dimensional analysis

1. a) Convert distance and time: 72 × 1,000/3,600 = 20 m/s.

2. c) One g/cm³ is 0.001 kg divided by 0.000001 m³, or 1,000 kg/m³. Multiply by 2.

3. b) Square the length conversion before multiplying: 3 × $(0.01)^2$ = 0.0003 m².

4. a) Convert 2 cm² to 0.0002 m². Then pressure = 100/0.0002 = 500,000 Pa.

5. d) The units of v^2/g are $(m^2/s^2)/(m/s^2)$ = m. In contrast, v/g has units s, vg has units m^2/s^3, and g/v^2 has units 1/m.

6. a) Multiply 0.5 by the stated 3.6 × 10^6 J per kWh to obtain 1.8 × 10^6 J.

Exercise 19 · Binary and hexadecimal number systems

1. b) The occupied places are 8, 4 and 1. Their sum is 13.

2. d) 26 = 16 + 8 + 2, so the binary digits are 11010.

3. d) The first digit contributes 2 × 16 and F contributes 15. The result is 47.

4. d) Pad to 0010 1000. The groups correspond to hexadecimal digits 2 and 8.

5. a) 1111 represents 15. Adding 1 gives 16, but modulo-16 storage discards the carry and stores 0.

6. b) Move each bit one position left and insert a zero on the right: 0011 becomes 0110.

Exercise 20 · Boolean logic and logic gates

1. c) B OR C is 1 because C is 1. Then A AND 1 is also 1.

2. b) An AND gate with one input equal to 0 produces 0. Therefore disabling A forces the indicator output to zero.

3. a) The first identity given in the passage is De Morgan’s relation. Both B and C must be zero for the negated OR to be one.

4. a) The inputs are equal, so XOR produces 0. OR would produce 1 for this same pair.

5. b) The distributive identity gives (A AND B) OR (A AND C): OR the outputs of the two AND gates. The other circuits differ on some input combinations.

6. a) A must be 1. Of the four B,C pairs, 01, 10 and 11 make B OR C equal 1. Thus exactly three combinations qualify.

Exercise 21 · Algorithms and step-by-step tracing

1. b) The array uses indices 0, 1, 2 and 3. The value 9 is at index 2.

2. d) The target is absent. The search checks all four elements before returning -1.

3. b) First swap 7 and 2. Next compare 7 and 9 without swapping. Finally swap 9 and 4, producing [2,7,4,9].

4. a) The decisions to discard a half rely on ascending order. The other listed properties are not required by the described routine.

5. a) The initial bounds are 0 and 3, so the middle index is floor(3/2) = 1. Its value is 4.

6. d) The routine stops at the first match. The first element has value 4 and index 0.

Exercise 22 · Algorithmic complexity and growth

1. b) In $3n + 5$, the linear term dominates for large n. Constant multipliers and additive constants do not change the tightest listed growth class.

2. d) The n^2 term dominates $n^2 + n$. Replacing n by 2n makes that dominant term four times as large.

3. c) $\log_2 8 = 3$, which is already an integer. Adding 1 gives four operations in this model.

4. b) For each of n outer iterations, the inner loop performs n operations. The total is $n \times n = n^2$.

5. d) Sequential work adds rather than multiplies. The n log n sorting term dominates the additional logarithmic search.

6. a) TA(1) = 8 and TB(1) = 2. A better asymptotic growth class does not guarantee a lower count for every small input.

Exercise 23 · Arrays, stacks and queues

1. b) B was pushed last and is at the top. The stack removes and returns B.

2. d) A entered first and is at the front. The queue removes and returns A.

3. a) Zero-based indices 0, 1, 2 and 3 correspond to values 9, 5, 7 and 1.

4. c) The first pop removes 5, leaving 3. Pushing 7 puts it on top, so the final pop returns 7.

5. d) A queue preserves arrival order for removal. A stack reverses the order of the most recently stored actions.

6. c) The first dequeue removes X, leaving the queue empty. The specified model reports underflow on another removal.

Exercise 24 · Recursion and iteration

1. d) Expand to 3 + 2 + 1 + F(0) = 3 + 2 + 1 + 1 = 7.

2. a) The nonnegative integer input decreases on each recursive step and reaches the explicitly defined base case.

3. c) Starting with 1, the procedure adds 1, then 2, then 3, producing 7.

4. c) The calls are F(4), F(3), F(2), F(1) and F(0), giving five in total.

5. b) G(5) = 2 + G(3) = 2 + 2 + G(1) = 5.

6. c) Only i = 1 and i = 2 are included. Starting from 1, the final value is 1 + 1 + 2 = 4.

Exercise 25 · Optimal order quantity and inventory costs

1. d) $Q^* = \sqrt{(2 \times 3{,}600 \times 25/2)} = \sqrt{90{,}000} = 300$ units.

2. b) Stock decreases steadily from 300 to zero. Its average over a replenishment cycle is $Q/2 = 150$ units.

3. c) Annual orders equal $D/Q = 3{,}600/300 = 12$.

4. a) S is inside the square root. Multiplying S by 2 multiplies Q^* by $\sqrt{2}$ rather than by 2.

5. a) Ordering cost is $3{,}600 \times 25/300 = €300$. Holding cost is $2 \times 300/2 = €300$. Their sum is €600.

6. c) A quantity discount makes purchase cost depend on order size. That conflicts with an explicit assumption and changes the objective that must be compared.

Exercise 26 · Break-even analysis

1. d) Subtract unit variable cost from price: 50 - 30 = €20.

2. b) Divide fixed cost by unit contribution: 12,000/20 = 600 units.

3. d) The total contribution is $20 \times 900 = €18{,}000$. Subtract €12,000 fixed cost to obtain €6,000 profit.

4. c) The new unit contribution is 55 - 30 = €25. Break-even quantity is 12,000/25 = 480.

5. d) The unit contribution falls to 50 - 35 = €15. Break-even quantity becomes 12,000/15 = 800.

6. a) Maximum contribution is $500 \times €20 = €10{,}000$, below €12,000 fixed cost. The resulting operating profit is -€2,000.

Exercise 27 · Accounting statements and transaction logic

1. a) Initial assets are €30,000. Subtract liabilities of €12,000 to obtain equity of €18,000.

2. b) Profit uses earned revenue, not only collected cash: €8,000 - €5,000 = €3,000.

3. a) The business earned €8,000 but collected €5,000, leaving €3,000 due from customers.

4. b) Cash is an asset and the repayment obligation is a liability. Borrowing does not itself earn revenue or create profit.

5. c) One asset increases by exactly the amount another decreases. Under the stated model, the purchase is not immediately expensed.

6. a) Depreciation reduces the equipment’s carrying value and period profit. With no other transactions, the expense also reduces equity; it is not a current cash payment.

Exercise 28 · Marketing mix, pricing and campaign comparisons

1. c) Durability is a feature of the item itself, so it is primarily a product decision.

2. d) Distribution and customer access fall under place. This differs from communicating an offer through advertising.

3. c) A produces 1,000 × 2% = 20 orders. Contribution is (20 - 8) × 20 - 150 = €90.

4. c) B produces 30 orders. Its contribution is (16 - 8) × 30 - 150 = €90, the same as A.

5. b) Before advertising, the campaign contributes (20 - 8) × 20 = €240. Spending exactly this amount on advertising leaves zero contribution.

6. b) Week and audience also changed. The arithmetic comparison is valid, but it does not isolate a causal price effect.

Exercise 29 · Project scheduling and the critical path

1. d) A finishes at 3, B at 7 and C at 5. D finishes at 12 and E at 7. F starts at 12 and finishes at 13.

2. a) A-B-D-F takes 3 + 4 + 5 + 1 = 13 days. The other valid listed paths are shorter; B is not a predecessor of E.

3. a) C then finishes at 6. B still controls D's start at 7, and E finishes by 8, before D. F still finishes at 13.

4. a) B finishes at 9, delaying D to finish at 14. F then finishes at 15.

5. c) D requires both B and C. Their finish times are 7 and 5, so D must wait until day 7.

6. d) E was already finished before the critical D branch. Shortening E does not change F's controlling predecessor or the 13-day duration.

Exercise 30 · Process capacity, bottlenecks and flow

1. d) The smallest stage capacity is assembly's 40 units/hour. Every unit must pass through it.

2. b) The capacities become 60, 55 and 50. Inspection becomes the bottleneck at 50 units/hour.

3. a) Demand is below the 40-unit line capacity. Actual sustained sales-supported output is therefore limited to 35 units/hour.

4. a) Assembly processes 35 units/hour out of its capacity of 40: 35/40 = 87.5%.

5. a) Convert 30 minutes to 0.5 hour. Little's law gives L = 40 × 0.5 = 20 units.

6. d) More waiting stock does not change assembly's 40-unit/hour processing limit. Each listed capacity increase would raise the original line limit, up to inspection's 50.

Exercise 31 · Supply, demand and market equilibrium

1. d) Set 100 - 2P = 20 + 2P. This gives 80 = 4P and P = 20.

2. d) At P = 20, demand is 100 - 40 = 60 and supply is 20 + 40 = 60.

3. b) Demand is 70 and supply is 50, giving a shortage of 20 units.

4. c) Set 120 - 2P = 20 + 2P. Then 100 = 4P and P = 25.

5. d) The new price is (100 - 40)/4 = 15. Quantity demanded at that price is 100 - 30 = 70.

6. c) At P = 10, Qd = 80 and Qs = 40. The shortage is 80 - 40 = 40 units.

Exercise 32 · Price elasticity and revenue

1. d) The quantity-change magnitude is 20/90. The price-change magnitude is 2/11. Their ratio is 11/9, approximately 1.22.

2. b) The midpoint elasticity is approximately 1.22, which exceeds 1.

3. b) Initial revenue is €1,000 and final revenue is €960. The change is -€40.

4. c) Use the local approximation: -0.5 × 10% = -5%.

5. b) With zero quantity response, elasticity is zero. On the stated axes, the demand curve is vertical.

6. d) At E = 1, the percentage quantity decrease approximately offsets the percentage price increase. This is a first-order statement, not an exact finite-change identity.

Exercise 33 · Cost curves and profit maximisation

1. d) Set 2 + 2q = 12, giving q = 5. Increasing marginal cost makes this the profit-maximising interior output.

2. b) TC(5) = 20 + 2 × 5 + 5² = €55.

3. b) Revenue is 12 × 5 = €60. Subtract total cost €55 to obtain profit €5.

4. a) The added fixed cost has zero derivative, so MC is unchanged. The output solving MC = MR remains 5 even though profit falls.

5. d) VC(5) = 10 + 25 = €35. Divide by five units to obtain €7 per unit.

6. d) Profit becomes 2q - (20 + 2q + q²) = -20 - q². This is largest at q = 0.

Exercise 34 · GDP, inflation and unemployment measures

1. d) Use year-2 quantity and year-2 price: 110 × €12 = €1,320.

2. b) Real GDP at the base price is 110 × €10 = €1,100.

3. d) Real GDP rises from €1,000 to €1,100. Growth is 100/1,000 = 10%.

4. c) The price index rises from 100 to 120, so inflation is 20%. This differs from nominal GDP growth.

5. d) The labour force is 920 + 80 = 1,000. The unemployment rate is 80/1,000 = 8%.

6. b) Unemployment becomes 60 and the labour force becomes 980. The rate falls to about 6.12%, while employment remains 920.

Exercise 35 · Market structures and monopoly decisions

1. d) The passage explicitly identifies firms under perfect competition as taking market price as given.

2. c) Price is 20 - q and marginal revenue is 20 - 2q. Their difference is q, so MR is lower whenever q > 0.

3. b) Set 20 - 2q = 4, giving 2q = 16 and q = 8.

4. d) Insert q = 8 into inverse demand: P = 20 - 8 = €12. The firm does not charge marginal revenue as its price.

5. c) Set P = 4 in P = 20 - q to obtain q = 16.

6. d) Strategic interdependence among a few important sellers is the defining feature of oligopoly in the passage.

Exercise 36 · Game theory and payoff matrices

1. b) Given Column’s C, Row receives 3 from C and 5 from D. Thus D is the best response.

2. a) Against C, D gives 5 rather than 3. Against D, D gives 1 rather than 0. D is better in both cases.

3. c) Both organisations prefer D against either opponent choice. At (D,D), switching alone to C lowers own payoff from 1 to 0.

4. d) The joint payoff is 6 at (C,C), 5 at either mixed-choice outcome, and 2 at (D,D). Mutual cooperation has the largest total, despite not being individually stable.

5. c) With Column still choosing D, the relevant cell becomes (C,D), whose Row payoff is 0.

6. b) Against C, defecting still yields 5 rather than 4. Against D, it still yields 1 rather than 0. Both organisations retain strictly dominant D.

Exercise 37 · Qualitative and quantitative research strategies

1. d) It starts with a hypothesis and tests a numerical relationship using standardised data, matching the stated quantitative, deductive approach.

2. a) Detailed accounts can reveal processes and conditions. Twelve deliberately selected interviews do not support the other population-level or experimental claims.

3. b) The passage allows an iterative qualitative process when changes are made transparent. Refinement does not itself establish prevalence or invalidate all prior data.

4. c) The issue is concealing the timing of hypothesis formation. Exploratory work can be useful, but should not be presented as a pre-specified confirmation.

5. b) Numerical records address patterns across cases, while interviews explore how those patterns could arise. The combination directly fits the two research aims.

6. b) Participation was not random. Selection and other differences may contribute to the association, so a causal effect is not established by this finding alone.

Exercise 38 · Sampling and statistical significance

1. b) The population is divided into strata, and random samples are selected within them. Equal sample sizes need not match population shares.

2. c) Weight by the population composition: $0.80 \times 0.40 + 0.20 \times 0.80 = 0.48$, or 48%.

3. b) The p-value describes a probability of data at least as extreme conditional on the null model. It is not a posterior probability of either hypothesis.

4. a) Because $0.03 < 0.05$, the pre-specified rule rejects the null. This decision does not establish certainty or justify changing the threshold.

5. a) The null asserts equal population rates. Rejecting it when that assertion is true is a Type I error.

6. c) A larger sample from the same selective source can retain systematic bias. Sample size alone does not repair an unrepresentative recruitment process.

Exercise 39 · Correlation, confounding and causal claims

1. b) A positive correlation describes an average direction of linear association. It does not imply universal ordering or a causal dose effect.

2. c) A common cause can create or contribute to an observed association between two other variables.

3. b) This explanation runs from an academic outcome or experience toward attendance, rather than assuming attendance is always the cause of performance.

4. b) Randomisation concerns the offer, so the clean comparison is between assignment to the offer and assignment to no offer.

5. a) Variables measured at the same time may leave temporal order unresolved, even when their association can be calculated.

6. b) The passage limits r to linear association. A zero value does not rule out a curved relationship or establish complete independence.

Exercise 40 · Survey and questionnaire design

1. c) A respondent may find the service reliable but expensive, or unreliable but cheap. One answer cannot clearly separate the two constructs.

2. c) “How would you rate the service’s punctuality?” asks for an assessment without supplying a positive or negative evaluation in advance.

3. c) The categories 0; 1-2; 3-4; 5 or more include every nonnegative whole-number count exactly once. The other sets either overlap, omit zero or omit counts above four.

4. a) A common explicit interval makes the frequency question more consistently interpretable across respondents.

5. a) Measurement changed along with time. An observed response difference may therefore mix a real trend with a wording effect.

6. d) Random assignment with other procedures held constant provides the clearest comparison of wording itself among the listed designs.

Exercise 41 · Argument analysis and logical validity

1. b) The premise says every approved design has passed this load test. The other conclusions go beyond that premise.

2. a) Passing is necessary but not stated to be sufficient. K could still fail another approval requirement.

3. d) If L were approved, it would have passed. Not passing therefore implies not approved, using the valid contrapositive pattern.

4. d) A counterexample to “passed implies approved” must satisfy passed while failing approved.

5. c) The reviewer substitutes a personal attack for examination of the estimate and its supporting reasons.

6. b) The criticism attacks a distorted, more extreme claim. The original proposal concerns unnecessary printing, not a ban on communication.

Exercise 42 · Social and psychological concepts in applied research

1. b) The message describes what others commonly do. It does not explicitly state what they approve of or provide a reward.

2. d) The message group has 300 returns among 500 students, giving 60%.

3. b) The rates are 60% and 40%. Their difference is 20 percentage points; the relative increase is a different quantity.

4. d) The target behaviour is returning a cup. An observed return count measures it directly, whereas the other measures concern attitudes or unrelated recognition.

5. c) Random assignment and the stated controls support a causal comparison of the message here. They do not establish universal generalisation or an exact mechanism.

6. b) A desirable consequence contingent on a behaviour that increases its future occurrence fits the stated definition of positive reinforcement.